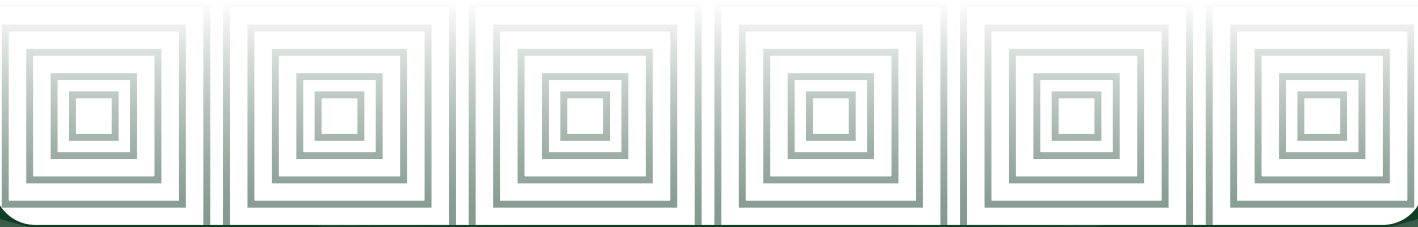


GEOGRAPHY SYLLABUS

FOR ADVANCED LEVEL (S4-S6)

Kigali, 2026



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FOREWORD

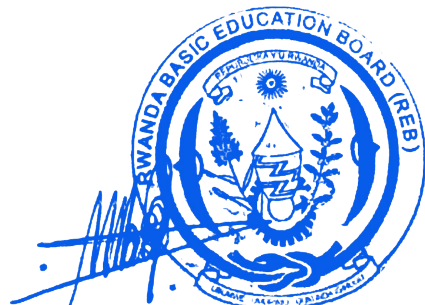
The Rwanda Basic Education Board (REB) is honoured to present the second edition of the Upper Secondary School Geography syllabus (S4–S6). It serves as an official guide for competence-based teaching and learning, ensuring consistency in delivering quality education across the country. Developed collaboratively by subject specialists, educators, and curriculum developers, it supports the implementation of the Competence-Based Curriculum (CBC) and reflects Rwanda's commitment to nurturing learners' knowledge, skills, values, and attitudes.

Quality education is the foundation of Rwanda's sustainable development and prosperity. Aligned with the national education philosophy, this syllabus promotes critical thinking, creativity, inclusiveness, and holistic learner development. It aims to empower young people with competencies necessary to achieve full potential, equipping them with relevant knowledge, skills, the right attitudes and values to integrate effectively into society and pursue opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the Second National Strategy for Transformation (NST2), this syllabus supports the nation's goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritizes learner-centred pedagogy, active learning, and the practical application of knowledge to prepare student not only for academic success but also to become responsible, ethical, and productive citizens capable of contributing to Rwanda's continued transformation.

This syllabus reflects our determination to equip learners with 21st-century skills, foster innovation, and instil values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our learners and supports Rwanda's national aspirations. Together, we can ensure that education continues to be a powerful driver of our nation's development and prosperity.



Dr. MBARUSHIMANA Nelson

Director General, REB

ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this Upper secondary school Geography syllabus (S4–S6).

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda's Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the Second National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda's education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among learners. Through your dedication, this syllabus will fulfil its purpose of empowering learners to realize their full potential and contribute meaningfully to Rwanda's sustainable development.



MURUNGI Joan

Head of Curriculum, Teaching and Learning Resources Department
Rwanda Basic Education Board (REB)

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1. GENERAL INTRODUCTION

1.1. Background to the syllabus adaptation

The adaptation of the geography syllabus for upper secondary school reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This second edition incorporates the Ministerial directive to harmonize school hours to five per day and 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as *Vision 2050* and the *Second National Strategy for Transformation (NST2)*, which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

Geography at the Upper secondary level aims to equip learners with foundational knowledge, practical skills, and positive values needed to identify opportunities, solve real-life problems, and contribute meaningfully to their communities. The subject encourages learners to think creatively, act ethically, and develop a sense of initiative, self-confidence, and social responsibility.

2. TEACHING AND LEARNING GEOGRAPHY

2.1. Rationale of Teaching and Learning Geography

Geography plays a central role in nurturing competent, innovative, and self-reliant learners who contribute to sustainable national development. It enables learners to understand the dynamics of economic life, recognize business opportunities, and apply creativity and problem-solving in diverse contexts.

Through this subject, learners develop an entrepreneurial mindset characterized by curiosity, initiative, resilience, and ethical responsibility. It encourages them to transform ideas into action and to approach challenges as

opportunities for innovation. In doing so, learners build both economic and social value, respect for work, aligning with Rwanda's aspiration to develop responsible citizens and job creators rather than job seekers.

2.1.1. Geography and Society

Geography helps people understand the close relationship between society and the physical environment in which they live. It equips students with knowledge about places, resources, cultures, and human activities, while enabling them to analyse social and environmental issues from different perspectives. Through geographical studies, students develop an understanding of how societies interact, depend on one another, and influence the environment. Geography also promotes sustainable development, responsible citizenship, and critical thinking, preparing students with the knowledge and skills needed for adult life, employment, and active participation in society.

2.1.2. Geography and Student

Geography provides students with opportunities to explore their surroundings, develop observation and problem-solving skills, and build awareness of how people interact with the natural world. It strengthens communication, teamwork, and critical thinking through field studies, mapping, and environmental projects.

Through practical activities and community-based learning, Geography connects theory to real-life situations, encouraging reflection, responsibility, and adaptability. It nurtures curiosity, environmental stewardship, and a strong commitment to contribute positively to families, communities, and national sustainable development.

2.2. Competences

A competence is the ability to use an appropriate combination of knowledge, skills, attitudes, values, and behaviours to accomplish a task successfully. In this syllabus, competences reflect a holistic blend of these elements, enabling learners to apply them confidently and appropriately in diverse real-life situations.

By acquiring these competences, learners become better equipped to handle challenges, make informed decisions, act responsibly, and contribute positively as active members of society. The competences are embedded in the broad subject competences, yearly objectives, and unit-specific learning competencies. Both generic competences and basic competences must be emphasized and consistently reflected in the teaching and learning process.

Teachers are encouraged to design learning experiences that engage learners in meaningful tasks, allowing them to practice and demonstrate competences both during and at the end of the learning process.

2.2.1. Generic competences

Generic competences are cross-cutting abilities developed across all subjects. They equip learners with essential lifelong learning skills and underpin all other competences in the curriculum.

- **Critical and problem-solving skills:** Learners develop the ability to think broadly and objectively, evaluate their causes and consequences, and suggest peaceful and practical solutions that contribute to harmony and sustainable living in their communities.
- **Creativity and innovation:** The acquisition of such skills will help learners to take the initiative and use imagination beyond the knowledge provided, to generate new ideas and construct new concepts.
- **Research:** This will help learners acquire the ability to gather, analyze, and interpret information from a variety of sources, including observations, interviews, and case studies to inform decisions and develop evidence-based conclusions.
- **Communication in official languages:** Teachers, irrespective of their subject, will ensure proper use of the language of instruction to help learners communicate clearly, confidently, and respectfully in both spoken and written forms, enabling them to express ideas, negotiate, and present information effectively for logical and coherent exchange.

- **Cooperation, interpersonal management, and life skills:** This competence enables learners to work effectively with others, respect diverse opinions, and show empathy in group activities. It fosters moral values, responsibility, and active participation in community-oriented tasks such as environmental protection, health promotion, and service. Learners also build social skills that enhance teamwork, leadership, and adaptability in addressing life's challenges.
- **Lifelong learning:** learners cultivate curiosity, initiative, and adaptability by linking classroom knowledge to real-life experiences. This empowers them with the ability to update knowledge and skills independently, cope with evolving societal needs, and pursue personal growth and continuous development.

2.2.2. Geography and broad competences

At the end of upper secondary education, learners should be able to demonstrate the following broad competences specific to Geography.

- Demonstrate a strong understanding of fundamental concepts in physical and human geography.
- Identify and describe major physical geographical phenomena and explain their influence on sustainable development.
- Analyse the interactions between physical environments and human activities, including human modifications of the environment.
- Appreciate the interdependence between physical and human aspects of geography in shaping global and local systems.
- Understand issues related to sustainable growth, conservation, reconstruction, and national development within regional and global contexts.
- Evaluate global and national socio-economic problems and propose realistic solutions.
- Apply geographical research methods, including accurate collection, handling, and interpretation of data.
- Read, analyse, and interpret maps, photographs, and other geographical information using appropriate techniques

- Develop essential cross-cutting skills such as critical thinking, problem solving, communication, cooperation, investigation, and effective presentation of conclusions.

2.2.3. Geography and Developing Competences

National policies emphasize developing competences that promote critical thinking and responsible citizenship through interactive and practical learning.

Geography supports this goal by engaging learners in collaborative activities such as projects and community tasks that build creativity, communication, teamwork, and ethical decision-making. By connecting classroom learning to real-life situations, it fosters innovation, confidence, and responsibility, preparing learners for employment, self-employment, and active community involvement.

3. PEDAGOGICAL APPROACH

The pedagogical approach in Geography under the CBC emphasizes learner-centered, participatory, and inclusive methods that develop knowledge, skills, values, and competences for responsible citizenship. Teaching is interactive and active, engaging learners through debates, discussions, role-plays, storytelling, field visits, projects, research, case studies, and reflective activities. These approaches ensure that learning is enjoyable, personalized, and connected to real-life situations, while also fostering critical thinking, moral reasoning, and problem-solving.

3.1. Role of the Teacher

The teaching and learning of Geography in upper secondary level must reflect the Competence-Based Curriculum (CBC) principles, which promote learner-centred, participatory, and practical methods. The main goal is to guide learners acquire and apply entrepreneurial knowledge, skills, values, and attitudes that prepare them to become innovative, responsible, and productive citizens.

Teachers should therefore move beyond traditional lecturing to interactive strategies that encourage inquiry, creativity, problem-solving, and collaboration. Learning must link theory with practice, making classroom experiences relevant to the learner's environment and everyday life.

3.2. Role of the learners

Learners are active participants and co-creators of knowledge. They engage in inquiry, debates, problem-solving, and group collaboration while drawing connections between personal experiences. Through these activities learners develop critical thinking, communication, respect for diversity, moral responsibility, and the ability to apply knowledge in everyday life and societal challenges.

3.3. Inclusive and Special Needs Education

In line with Rwanda's inclusive education policy, all learners regardless of ability have the right to access quality education. Teachers adapt teaching methods, materials, and assessment strategies to meet diverse needs, including those of learners with disabilities. Differentiated instruction, assistive technologies, peer support, and individualized approaches are used to ensure equitable participation and achievement of learning outcomes. This inclusive approach recognizes learners' varied paces, experiences, and abilities, creating an environment where every learner can thrive.

4. ASSESSMENT APPROACH

Assessment in the Competence-Based Curriculum (CBC) is an essential part of the teaching and learning process, designed to evaluate learners' progress in knowledge, skills, values, and attitudes while supporting continuous growth. It emphasizes real-life application and problem-solving, enabling learners to demonstrate their competences in authentic contexts. Assessment involves systematically gathering and interpreting evidence to measure performance against established standards, providing constructive feedback to guide both teaching and learning.

In Geography, assessment focuses on learners 'ability to demonstrate creativity, innovation, problem-solving, ethical behaviour, and teamwork. It encourages reflection, continuous improvement, and the development of responsible entrepreneurial mindsets.

Assessment should therefore be comprehensive, covering both formative (continuous) and summative (end-of-term or national) evaluations. It should be transparent, fair, inclusive, and competence aligned.

4.1. Levels of assessment in advanced level across terms and national examinations

To provide a comprehensive picture of learner's achievement and development, assessment is conducted at multiple levels:

- i) **Diagnostic:** Conducted before instruction to identify learners 'prior knowledge and skills; administered at the beginning of a unit or lesson to guide teaching.
- ii) **Formative / Continuous:** Carried out during the learning process to track progress and provide feedback; implemented continuously through quizzes, projects, class activities, and teacher observations.
- iii) **Summative assessment:**
 - **End term I:** Administered after instruction to evaluate learners 'achievement of Term I content; conducted at the end of the first term as a school-based assessment.
 - **End term II:** District-coordinated assessment designed to measure achievement of Term II content; developed by schools and quality-checked at district level.
 - **End term III:** National assessment to evaluate mastery of the entire year's content; prepared, set, and moderated by NESA at the end of the third term.
 - **National examinations:** High-stakes assessments for certification, selection, and accountability; are administered nationally at the end of key education cycles (P6, S3, S6) under NESA's authority. This examination integrates objective questions, structured responses, and applied tasks aligned with CBC principles to holistically measure learner achievement.

- In advanced level, all students will take 11 subjects in total -7 core subjects and 4 compulsory subjects. The compulsory subjects include ICT, Entrepreneurship, General and religious education, and Physical education and sport, which are mandatory for all learners across all pathways.
- All subjects will be assessed at the national level, except for Physical Education and Sport, General and Religious Education, ICT, and Entrepreneurship, which will be assessed at the school level only.

Average scores from school, district, and comprehensive summative assessments will be weighted and included in the final national examination grading, initially contributing 10%, with the percentage gradually increasing as teachers gain experience and confidence in assessment practices.

4.2. Record keeping

Accurate record keeping is essential for monitoring learner progress and informing instructional decisions. Teachers use tools such as portfolios, checklists, and report books to collect evidence of learning. Portfolios include samples of learners' work, test results, projects, and self-reflections, serving as both a record of achievement and a diagnostic tool for guiding remedial action. These records provide data for feedback to learners, teachers, and parents, and contribute to final evaluations.

4.3. Item writing in summative assessment

Summative assessment items must be valid, fair, reliable, and aligned with CBC principles. They should test a range of competencies, including knowledge recall, application, analysis, and values integration, while reflecting real-life contexts. Item writers follow a structured process that involves:

- Identify topic areas to be tested from the subject syllabus.
- Outline subject matter content to be considered as the basis for the test.
- Identify learning outcomes to be measured by the test.

- Preparing a test blueprint or table of specifications;
- Using verbs that require higher-order thinking beyond memorization;
- Applying transparent and consistent marking schemes.
- Greater weight is given to higher-order cognitive skills as stated in the syllabus.

4.4. Structure and format of national examinations

At the completion of upper secondary school education, learners undertake a national examination designed to evaluate literacy, numeracy, and competences in core subjects. For Geography, a single combined paper is administered, integrating both subject areas. The examination employs a variety of item types such as objective questions, structured responses, and applied tasks intended to holistically assess both subject knowledge and broader CBC competencies.

4.5. Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent-teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where learners are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well, meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports learners' continued growth.

5. Resources

The effective implementation of the geography syllabus requires the provision and utilization of appropriate resources. These resources include both teaching and learning materials and qualified human personnel, without which the delivery of the syllabus would be ineffective.

Teachers are encouraged to utilize both locally available and digital resources to create rich, relevant, and engaging learning experiences that promote creativity, problem-solving, and entrepreneurial thinking.

5.1. Materials needed for implementation of this syllabus

A variety of materials are essential to facilitate teaching and learning processes in Geography. These materials provide learners with practical experiences, stimulate critical thinking, and foster the acquisition of competences. They include:

- **Textbooks and reference materials:** Updated textbooks, teacher's guides, learners' books, dictionaries, and business-related reference materials that support theoretical and practical learning.
- **Teaching and learning aids:** Visual and audio-visual resources such as maps, charts, posters, diagrams, models, globes, projectors, and videos, as well as digital tools like e-books, educational websites, and online platforms, are used to illustrate geographical features, environmental processes, and human-environment interactions, and to enhance learners' understanding of the world around them.
- **Library and resource centre materials:** Access to a well-equipped library with supplementary reading materials such as journals, case studies, and stories of Rwandan entrepreneurs for extended reading and inspiration.
- **Geography and simulation resources:** maps, globes, weather tools, and environmental games for activities like map reading, weather observation, resource use, and disaster preparedness.
- **Field visit resources:** Materials to support educational tours and fieldwork, such as journals, notebooks, and logistical support, since field visits and community interactions are integral to experiential learning in this subject.

- **Community and cultural resources:** Access to local leaders, religious institutions, museums, cultural heritage sites and genocide memorial sites that provide learners with real-life exposure to the values and practices being studied.
- **Assessment tools:** Rubrics, checklists, portfolios, and test materials are necessary to support continuous and summative assessment in line with the competence-based approach.

These resources should be adapted to the learners' context, ensuring inclusivity and accessibility for learners with special educational needs.

5.2. Human resource

The successful implementation of this syllabus relies heavily on qualified, competent, and motivated human resources. The following are key personnel required:

- **Teachers:** Professional and well-trained teachers with subject expertise and strong pedagogical skills in learner-centered, participatory, and inclusive methods. Teachers should model integrity, tolerance, respect, responsibility, facilitate projects and apply effective assessment strategies.
- **Community resource persons:** Local entrepreneurs, cooperative members, financial institutions, and professionals who share real-life experiences and act as mentors during classroom sessions or field visits.
- **Parents and guardians:** They are partners in education who provide moral support, monitor learners progress, and collaborate with teachers to reinforce entrepreneurial values, creativity, responsibility, and other competences at home, ensuring holistic learner development.
- **Local authorities and partners:** Support schools through linkages with business development services, innovation hubs, and community-based projects that offer practical learning opportunities.

Effective collaboration among stakeholders creates an enabling learning environment that equips learners with geographical knowledge, values, and practical competences needed for responsible citizenship, ethical behaviour, and meaningful contribution to national development.

5.3. Skills required for the Teacher of Geography

Effective implementation of this syllabus requires teachers with strong subject knowledge, pedagogical skills, and interpersonal competence. As facilitators and role models, they guide learners to develop geographical mindsets and apply competences in real-life contexts. To achieve this, teachers should demonstrate the following key skills:

a) Subject matter expertise

- Strong mastery of geography concepts such as landforms, climate, population, natural resources, and sustainable development.
- Ability to link geographical theory with real-life examples and national development priorities, including *Vision 2050* and *NST2*.
- Skill in relating Geography to other disciplines such as *Mathematics*, *ICT*, *Economics*, and *Social Studies*.
- Awareness of current trends and technologies that influence *environmental management*, *climate change adaptation*, and *sustainable development* locally and globally.

b) Pedagogical and instructional skills

- Ability to design and implement learner-centred and participatory lessons that encourage active engagement, creativity, and critical thinking.
- Use of diverse teaching methods such as brainstorming, project-based learning, role-playing, case studies, and community engagement.
- Creativity and innovation in developing and applying diverse teaching resources, including ICT tools, realia, and community-based materials.
- Ability to adjust instruction to learners' levels, ensuring inclusivity and engagement for all.

c) Assessment skills

- Proficiency in designing and applying formative and summative assessments aligned with curriculum competencies.
- Effective use of rubrics, portfolios, checklists, and peer assessments to evaluate learners' competences and provide feedback that motivates learners and guides improvement.
- Skill in developing valid and fair test items that assess higher-order thinking and practical skills.
- Ability to analyze assessment results and use them to guide remedial or enrichment activities.

d) Classroom management and facilitation

- Ability to establish a respectful, safe, and inclusive learning environment that promotes tolerance and values diversity.
- Proficiency in managing group work, discussions, and collaborative learning activities.
- Consistent modelling of positive behaviour, respect, punctuality, and integrity as an example for learners.

e) Communication and interpersonal skills

- Strong oral and written communication to simplify complex ideas, stimulate dialogue, and foster critical reflection.
- Active listening and empathy to respond effectively to learners' academic and personal needs.
- Collaboration with colleagues, parents, school leaders, and community stakeholders to support holistic learners' growth.
- Ability to motivate and inspire learners through encouragement and positive reinforcement.

f) Guidance and counselling skills

- Ability to provide moral, emotional, and motivational guidance to learners to build confidence, resilience, and ethical conduct.
- Demonstrating passion, patience, impartiality, and genuine care in dealing with learners' individual differences and challenges.
- Supporting learners in exploring business ideas, career paths, and responsible life choices.

g) Professional and ethical competence

- Upholding high ethical standards, integrity, impartiality in all teaching and assessment practices, and ethical conduct, serving as a role model for learners.
- Capacity to balance academic instruction with the cultivation of values and responsible citizenship.
- Demonstrating commitment to continuous professional development (CPD) through reflective practice, peer collaboration, and research.
- Modelling Geography values such as initiative, responsibility, and self-discipline.

h) Research and innovation skills

- Engagement in reflective practice and classroom-based research to improve teaching and learning.
- Adaptation of innovative approaches and new technologies to enhance learner participation.
- Connecting global Geography trends to Rwanda's socio-economic realities.
- Encouraging learners to use research findings in designing projects and solving community challenges.

The teachers who possess these competences will not only deliver Geography effectively but will also serve as role models who inspire learners to become environmentally conscious, analytical, and responsible contributors to Rwanda's sustainable development.

6. SYLLABUS UNITS

6.1. Presentation of the structure of the syllabus

The geography syllabus is designed as a single, integrated syllabus implemented uniformly across all grades. Its structure follows a logical progression that ensures clarity, consistency, and alignment with the Competence-Based Curriculum approach.

At each grade level, the syllabus is organized into **Topic Areas**, which are further broken down into **Sub-Topic Areas**. Each Sub-Topic Area is composed of Units, and every unit is characterized by the following features:

1. **Alignment with Lessons:** Each unit corresponds to a specified number of lessons, ensuring effective time allocation and instructional planning.
2. **Key Unit Competence:** Every unit is built around a Key Unit Competence that guides all teaching and learning activities. This competence represents the overall learning goal to be achieved by both teacher and learner.
3. **Learning Objectives:** Each key unit competence is translated into specific learning objectives, categorized as follows:
 - **Type I (Knowledge and Understanding):** Focused on foundational knowledge and comprehension, often referred to as *Lower Order Thinking Skills (LOTS)*.
 - **Type II and Type III (Skills, Attitudes, and Values):** Aimed at developing practical skills, critical thinking, ethical values, and positive attitudes. These objectives, known as *Higher Order Thinking Skills (HOTS)*, form the core focus of the adapted Competence-Based Curriculum.
4. **Content Specification:** Each unit outlines the scope of content to be taught and learned, ensuring that teachers and learners work within the boundaries set by the stated learning objectives.

SENIOR FOUR

6.2. GEOGRAPHY FOR SENIOR 4

6.2.1. General competences of geography at the end of Senior 4:

At the end of senior four a learner will be able to:

1. Demonstrate an understanding of the earth and the elements of physical geography.
2. Explain the relationship between physical geography and human activities.
3. Apply knowledge and understanding of physical and human geography theory.
4. Read and identify the elements of a map and interpret them.

6.2.2. Syllabus units for senior four

Unit	Unit title	Periods
1	Field work Techniques	18
2	Maps and grid reference systems	12
3	Formation of relief features in Rwanda	28
4	Rocks in Rwanda	5
5	Soils in Rwanda	8
6	Climate in Rwanda	10
7	Vegetation in Rwanda	8
8	Drainage system in Rwanda	9
9	Population in Rwanda	12
10	Rural and urban Settlement in Rwanda	24
11	Agriculture in Rwanda	26
12	Forestry in Rwanda	12
13	Fishing in Rwanda	8
14	Mining in Rwanda	7
15	Power and Energy in Rwanda	10
16	Industry in Rwanda	12
17	Transport, Communication and Trade in Rwanda	12
18	UNIT 18: Environmental Conservation in Rwanda and Tourism	10
Total		231

GEOGRAPHY SYLLABUS FOR SENIOR FOUR

Topic Area: Practical Geography		Sub-Topic Area: Field work and statistics		
S4 Geography		Unit 1: Field work Techniques		No. of Periods: 18
Key unit competence: Interpret the geographical data gathered during fieldwork				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of the concept of a field work. – Identify the various methods used in the data collection during the field work. – State the advantages and disadvantages of each method of a field work.	– Describe the various methods used in collecting the data during the field work. – Evaluate the advantages and disadvantages of various methods of a field work. – Analyse the field work procedures from the preparation stage to the reporting stage.	– Show concern for the problems affecting our society and carry out a field study to find the causes and the solutions. – Show respect for the protection of the natural and man-made features in our environment.	– Meaning of a field work – Field work methods (observation, questionnaire, interview, recording, sampling and measuring) – Advantages and disadvantages of each field work method. – Field work procedures/ stages (pre-field preparation, pilot study, actual fieldwork/data collection stage and follow up/post field).	– Use the geographical documents, internet, to identify the field work methods, the advantages and disadvantages of each method, field work procedures and the presentation of the field work findings. – Write down the findings.

– State the field work procedures from the preparation stage to the reporting stage. – Identify the problems affecting the planning and the implementation of a field work and their solutions.	– Explain the problems affecting the planning and the implementation of a field work and their solutions.		– Field work case studies: school area, section of a river valley, landform in an area, urban area, settlements, market area, fishing industry and plantations. – Problems affecting the planning and the implementation of a field work. – Solutions to the problems affecting the planning and the implementation	– In groups, discuss and compile the findings for a class presentation followed by question and answers. – Record the findings about the field work techniques for the discussions in reference to the knowledge acquired in the current unit. – Draw relevant conclusions in a report.
Links to other subjects: <i>Biology: Studying ecosystems.</i>				
Assessment criteria: <i>The learners are able to correctly interpret the geographical data gathered from a conducted fieldwork</i>				
Teaching and learning materials: <i>Geographical documents, maps, cameras, radio recorder, photographs, local environment, tape measures, GPS, talking and tactile materials, objects and internet.</i>				

Topic Area: Practical Geography		Sub-Topic Area: Map work interpretation		
S4 Geography		Unit 2: Maps and Greed reference systems		No. of periods: 12
Key unit competence: Describe different types of maps and accurately use grid reference systems to locate geographical features on maps.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Outline the elements of a map. – Identify different types of maps. – Identify the categories of grid lines used to locate phenomena on a map	– Classify different maps according to the content and scales. – Locate the phenomena on the topographic maps using grid reference systems.	– Appreciate the importance of using maps in everyday life – Appreciate the importance of grid references for locating phenomena or places on the maps.	– Meaning and elements of a map Types and examples of maps: According to the content: – Topographic maps and thematic maps (e.g: a political maps) – According to the scale: small scale map, medium scale map, large scale map and a plan	– Use different maps to identify the elements of map and write down the findings. – In groups, discuss the findings for a class presentation. – Use the geographical documents, maps and internet to identify different types of maps according to the content and scale. Write down the findings. – In groups, discuss the findings for a class presentation, engage in question-and-answer methodology.

			– Location of the phenomena (Grid reference system) on the maps.	– Use the grid lines (4- and 6-figure grid references) to locate a point /a place or phenomena on the maps using.
Links to other subjects: <i>Mathematics: Constructing graphs (x and y axis) coordinates</i>				
Assessment criteria: <i>The learners are able to correctly distinguish different types of cartographic projections and types of maps</i>				
Teaching and learning materials: <i>Geographical documents, internet, Globe, tactile and talking materials, topographic maps, drawing kit, graph papers and local environment.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Landform evolution and processes		
S4 Geography		Unit 3: Formation of relief features in Rwanda		No. of periods: 28
Key unit competence: Examine the formation processes of major relief regions of Rwanda and their effects on human activities.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– State the area of Rwanda in terms of land and water surface area. – Describe the population composition – Outline the administrative divisions of Rwanda. – Identify the major relief regions of Rwanda	– Locate Rwanda on the map of Africa. – Draw a sketch map of Rwanda locating its administrative divisions. – Distinguish the major relief regions of Rwanda on the map.	– Appreciate the natural beauty of Rwanda. – Contribute to the national policies of maintaining a healthy population and safe environment in the country. – Appreciate the importance of different relief features of Rwanda.	General presentation of Rwanda: – Location (geographical coordinates and the neighbouring countries). – Size (land surface, water surface, regional and international comparison). – Administrative divisions (capital city, provinces, districts and sectors). – Population (Number, life expectancy, standard of living). – The relief regions of Rwanda: Bugarama plains and rift valley, Central plateaus, Eastern plains, Buberuka Highlands, Volcanic region and Congo-Nile crest	– Use a map of Africa to locate Rwanda and write down the geographical coordinates and the neighbouring countries. – In groups, discuss the problems related to the geographical location of Rwanda. – Write down the findings from the group discussions and make a class presentation.

– Outline the endogenic and exogenic processes leading to the formation of different relief features of Rwanda. – Give the importance of the relief features to Rwanda. – Identify the problems related to the relief features of Rwanda.	– Describe the relief features of Rwanda shaped by internal and external geomorphic processes – Evaluate the importance of different relief features to Rwanda. – Examine effects of the running water, river erosion and deposition, wave erosion and deposition, mines and quarries on the landscape of Rwanda.	– Show continual desire to minimise the dangers of soil erosion.	Geomorphologic processes Endogenic/internal/tectonic processes: – The process of folding and the associated landforms (fold mountains, basins, spurs, anticlines, synclines, mountain ridges, waterfalls and valleys) in Rwanda. – The process of faulting and the associated landforms (block mountains, scarp slopes/ escarpments, graben /rift valleys, rift valley lakes, waterfalls, and fault guided valleys) in Rwanda – The process of volcanicity and the associated landforms (volcanoes, craters, crater lakes, caldera, caldera lakes, lava plains, lava plateaus, lava domes, waterfalls and lava dammed lakes) in Rwanda – The process of earthquake and its associated consequences in Rwanda	– Use a map of Rwanda to locate and identify the administrative divisions. – Conduct research using geographical documents, journals and internet to prepare a class presentation on the population size, composition, standard of living and life expectancy in Rwanda. – Use a drawn sketch map and locate the major relief regions of Rwanda.
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			– Warping (down and up warping) and its associated landforms (basins, basin lakes and plateaus) in Rwanda Exogenic processes: – Weathering and the resultant features (inselbergs, scars and tors) in Rwanda. – Mass wasting (landslides, mudflow, slumping): location, causes and consequences in Rwanda – River erosion and associated landforms (waterfall, rapids, gorges, canyons potholes, plunge pools, river cliffs, bluffs, spurs and V-shaped valleys) – River deposition and associated landforms (meanders, ox-bow lakes, flood plains, U-shaped valleys, alluvial fans and levees) – Wave erosion and associated landforms (cliffs, bays, headlands, and Notches / Wave-cut platforms, stack, caves, blowholes and arches)	– Use the geographical documents and internet to describe the processes /mode of formation of the various relief features of Rwanda. – In a group work, discuss the importance of different relief features of Rwanda and present the findings to the class. – In a field work study, find out the areas affected by the soil erosion and weathering, and write a report of the findings related to the landscape of the visited area.
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			– Wave deposition and associated landforms (tombolo, spits, beaches, mud plains and bars) – The importance of relief features on the development of Rwanda. – Environmental problems resulting from endogenic and exogenic processes in Rwanda	– In pairs, using evidence gained about the relief of Rwanda and the processes of weathering and erosion, evaluate how this has affected the population distribution and their activities in Rwanda – Write your findings as a short essay.
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Links to other subjects: *Economics: Chemistry: Chemical reactions (oxidation, carbonation, hydrolysis, solution and hydration).*

Assessment criteria: *The learners are able to accurately examine the formation processes of major relief regions of Rwanda and their effects on human activities*

Teaching and learning materials: *Geographical documents, journals, maps, photographs, local environment, concrete objects, recorders, internet, statistical and fieldwork materials and equipment. Talking and tactile materials like tactile maps, smart talking globes, smart tactile atlas and brailled textbooks.*

Topic Area: Physical Geography		Sub-Topic Area: Rocks and Minerals		
S4 Geography		Unit 4: Rocks in Rwanda		No. of periods: 5
Key unit competence: Analyze the characteristics of different types of rocks and their uses in Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the different types of rocks in Rwanda. – Locate the major rocks in Rwanda. – Identify the economic importance of rocks and minerals. – Identify the negative effects of rocks on human activities	– Explain the physical properties of the rocks. – Analyse characteristics of different types of rocks found in Rwanda. – Draw a sketch map of Rwanda to locate the major types of rocks. – Evaluate the importance of rocks in the development of Rwanda	– Show sustained efforts to exploit rocks without harming the environment – Appreciate the importance of rocks in Rwanda.	– Types, characteristics and location of rocks in Rwanda – Importance of rocks in Rwanda – Negative effects of rocks on human activities in Rwanda	– Conduct a field trip around the school to gather information on the types and characteristics of local rocks. Write down the findings. In groups, discuss the findings and make a class presentation. – Use the geographical documents and internet to research on the classification and characteristics of rocks in Rwanda. Record the findings and present them to the class.

	– Examine the negative effects of rocks on human activities			– Use the internet and textbooks to research the uses of rocks. Record and present your findings to the class. – Use the internet and textbooks to research the negative effects of rocks on human activities. Record and present your findings to the class.
Links to other subjects: <i>Chemistry: chemical properties.</i>				
Assessment criteria: <i>The learners are able to critically analyze the characteristics of different types of rocks and their uses in Rwanda</i>				
Materials: <i>Text books/ brailled textbooks, maps, atlases, tactile and talking materials, photographs, internet/ Jaws software, local environment, rock and mineral samples.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Soils in Rwanda		
S4 Geography		Unit 5: Soils in Rwanda		No. of periods: 8
Key unit competence: Examine the relationship between different soil types and human activities in Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify different types of soils and their characteristics in Rwanda. – Outline the causes and the effects of soil erosion in Rwanda. – Identify the importance of soils in Rwanda.	– Analyse characteristics of different types of soils in Rwanda and their location. – Explain causes and effects of soil erosion and soil conservation. – Evaluate the importance of soils in Rwanda.	– Appreciate the value of different types of soils in Rwanda. – Demonstrate a continuous commitment to propose mitigation measures for soil erosion in the community	– Types, characteristics and location of soils in Rwanda Soil erosion in Rwanda: – Types and causes of soil erosion in Rwanda – Effects of soil erosion in Rwanda – Soil conservation and soil management measures in Rwanda. – Importance of soil in Rwanda	– Use the map of Rwanda to locate different types of soils in Rwanda and write down the findings. – In groups, discuss the findings and make a class presentation supported by interactive question and answer. – Use geographical documents and internet to identify the causes and the effects of soil erosion and their conservation measures in Rwanda. Write down the findings. – In groups, discuss the findings and make class presentation.

				– In groups, discuss and write down the importance of soils in Rwanda and then make a presentation to the class.
Links to other subjects: <i>Biology: Study of ecosystems</i>				
Assessment criteria: <i>The learners are able to critically examine the relationship between different soil types and human activities in Rwanda</i>				
Teaching and Learning materials: <i>Geographical documents, maps, tactile and talking materials, photographs, internet, local environment, rocks and minerals samples.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Weather and Climate		
S4 Geography		Unit 6: Climate in Rwanda		No. of periods: 10
Key unit competence: Analyze the characteristics of climate of Rwanda and its impact on human activities				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the various Rwandan climatic regions and their characteristics. – Outline the factors influencing the climate in Rwanda. – Explain the relationship between the climate and human activities in Rwanda.	– Use a map of Rwanda to locate the different climatic regions of Rwanda. – Analyse the factors influencing the climate in Rwanda. – Analyse different seasons and their variations in Rwanda.	– Appreciate the importance of Rwandan climate on human activities – Show continual desire to protect the environment.	– Characteristics of the climate of Rwanda – Climatic zones of Rwanda – Factors that influence the climate of Rwanda (Altitude, latitude, vegetation, presence of water bodies, influence of winds, human activities). – Seasons in Rwanda: dry and wet seasons	– Use the map of Rwanda to locate different climatic regions of Rwanda – In small groups, discuss the reasons for different climatic regions in Rwanda and make a class presentation. – In groups, use the geographical documents and internet to describe the factors responsible for climatic and seasonal variations. Make a class presentation on the findings, supported by questions and answers.

	– Describe the relationship between the climate and human activities.		– Relationship between the climate and human activities	– In groups, discuss the relationship between the climate and human activities and make a class presentation.
Links to other subject: <i>Mathematics: Calculating mean and range using statistical techniques</i>				
Assessment criteria: <i>The learners are able to accurately analyze the characteristics of climate of Rwanda and its impact on human activities</i>				
Teaching and learning materials: <i>Geographical documents, brailled textbooks, maps, atlas, tactile and talking materials, photographs and local environment.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Vegetation		
S4 Geography		Unit 7: Vegetation in Rwanda		No. of periods: 8
Key unit competence: Examine the importance and methods to conserve different types of vegetation in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of vegetation – Outline different types of vegetation. – Explain the importance of vegetation. – Identify the factors influencing vegetation distribution. – Outline the causes and effects of over-exploitation of forests.	– Describe the various types of vegetation. – Examine the importance of vegetation – Describe the factors influencing vegetation distribution in Rwanda. – Analyse the causes and effects associated with over-exploitation of forests in Rwanda	– Appreciate the importance of vegetation to human activities. – Show continual desire to conserve and protect vegetation.	– Types of vegetation in Rwanda (natural and planted). – Distribution of natural forests in Rwanda – Factors influencing the types of vegetation – Importance of vegetation. – Causes and effects of over-exploitation of forests in Rwanda	– Use the geographical documents and internet to find out the types of vegetation, importance of vegetation and the factors influencing the vegetation distribution in Rwanda. Write down the findings. – In groups, discuss and compile the findings and then make a class presentation.

– Identify the measures of conserving vegetation.	– Describe the strategies / measures for vegetation conservation and protection.		– The conservation measures of vegetation	– Conduct a field study around the school on areas with severe forest exploitation; analyse the causes and effects, propose conservation measures, and draw conclusions for class presentation
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Links to other subjects: *Biology: Ecosystems and biomes*

Assessment criteria: *The learners are able to critically examine the importance and methods to conserve different types of vegetation in Rwanda*

Teaching and learning materials: *Maps, photographs, atlas, textbooks, documentary, diagrams, brailled textbooks, local environment, tactile and talking materials.*

Topic Area: Physical Geography		Sub-Topic Area: Drainage		
S4 Geography		Unit 8: Drainage system in Rwanda		No. of periods: 9
Key unit competence: Examine the relationship between drainage system of Rwanda and human activities.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– List the major rivers, lakes and swamps in Rwanda. – Explain the relationship between the drainage system and the human activities in Rwanda.	– Draw the sketch map of Rwanda and locate and label the major water bodies in Rwanda. – Explain the mode of formation of lakes in Rwanda. – Evaluate the importance of water bodies in the development of the human activities in Rwanda. – Analyse the Causes and effects of water pollution in Rwandan rivers and lakes	– Appreciate the importance of water bodies in Rwanda. – Advocate for the need to conserve the water bodies. – Appreciate the importance of water resource management and conservation.	– Major rivers and the drainage basins – Major lakes and their mode of formation – Major wetlands / swamps in Rwanda (Kamiranzovu, Rugezi, along the rivers like Nyabarongo, Akanyaru, Akagera). – Pollution of rivers, lakes, and wetlands: Causes and effects – Measures to promote the sustainable use of wetlands, lakes and rivers	– Use textbooks, internet, maps, atlases and photographs locate the major water bodies of Rwanda (rivers, lakes and swamps). In groups, discuss and write down the findings for a class presentation. – In groups, discuss the relationship between the drainage system and the human activities, then compile the findings for a class presentation.

	– Propose measures to promote the sustainable use of wetlands, lakes and rivers – Examine the relationship between drainage system and the human activities in Rwanda.		– Relationships between the drainage system (wetlands, lakes and rivers) and human activities in Rwanda	– In groups, discuss the causes and effects of water pollution, and propose the measures to promote the sustainable use of water bodies in Rwanda. Prepare the class presentation.
Links to other subjects: <i>Mathematics: Calculation of area, velocity, and volume</i>				
Assessment criteria: <i>The learners are able y to critically examine the relationship between drainage system of Rwanda and human activities</i>				
Teaching and learning materials: <i>Textbooks, internet, illustrations/diagrams, maps, photographs, slide shows, video clips, local environment, tactile and talking materials.</i>				

Topic Area: Human and Economic Geography		Sub-Topic Area: Population		
S4 Geography		Unit 9: Population in Rwanda		No. of periods: 12
Key unit competence: Analyse demographic problems in Rwanda and their possible solutions				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the key concepts related to population – Identify areas of high and low population density in Rwanda – Identify the causes of the rapid population growth and its effects in Rwanda.	– Analyse the factors influencing population distribution and its impact on the resource exploitation in Rwanda. – Illustrate the population structure and growth in Rwanda – Examine the causes and effects of rapid population growth and propose ways to control it	– Show concern about the causes and effects of population explosion on the sustainable development. – Take part in controlling the growth of the population.	– Meaning of population concepts (birth rate, death rate, growth rate, fertility rate, life expectancy). Population of Rwanda: – Population distribution and population density in Rwanda – Factors influencing population distribution in Rwanda. – Population structure of Rwanda.	– Use a population map of Rwanda, documents, internet to identify the areas of high and low population density. – In groups, discuss the factors influencing population distribution in Rwanda and make a class presentation. – Use the geographical documents and internet analyse the population structure and the factors for the population growth in Rwanda. Write and compile the findings for a class presentation.

– Identify the types, causes, and effects of migration in Rwanda	– Evaluate the causes and the effects of migration.		– Population growth in Rwanda. – Factors influencing the population growth in Rwanda. – Effects of the population growth in Rwanda. – Solutions for the rapid population growth. – Migrations: types, causes and consequences	– Use the geographical documents, internet or the to evaluate the consequences and the possible solutions for the rapid population growth. In groups, discuss and compile the findings for class presentation. – In pairs, examine the causes and consequences of population migration.
Links to other subjects: <i>Mathematics: Statistics.</i>				
Assessment criteria: <i>The learners are able to critically analyse the demographic problems in Rwanda and their possible solutions</i>				
Teaching and learning materials: <i>Geographical documents, internet, illustrations/diagrams, maps, photographs, slide shows, video clips, tactile and talking materials and local environment.</i>				

Topic Area: Human and Economic Geography		Sub-Topic Area: Settlement and urbanisation		
S4 Geography		Unit 10: Rural and urban settlement in Rwanda		No. of periods: 24
Key unit competence: Examine the impact of rural and urban settlements on sustainable development in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Outline the types of rural settlement – Identify the factors and effects of rural settlement. – Identify the characteristics and functions of the urban centres in Rwanda.	– Analyse the types and characteristics of rural settlement. – Examine the effects of rural settlements and propose possible solutions – Examine the reasons to implement the policy of rural settlement schemes (imidugudu) – Analyse the characteristics and the functions of the urban centres in Rwanda.	– Appreciate the importance of rural settlement schemes in Rwanda. – Appreciate the role of the urban centres in development of Rwanda.	Rural settlements: – Types and characteristics of rural settlements. – Factors influencing rural settlements – Problems and solutions of rural settlements. – Government policy towards rural settlement. Urban settlement – Characteristics of urban centres.	– Use the geographical documents, internet and photographs identify the types, characteristics, factors, problems and solutions of the rural settlement in Rwanda. In groups, discuss and compile the findings for a class presentation. – Use the geographical documents, internet and photographs to describe characteristics and functions of major urban centers in Rwanda.

– List the factors favouring the growth of the urban centres in Rwanda. – Identify the major urban centres of Rwanda. – Identify the impacts of urbanization in Rwanda	– Evaluate the factors for the growth of the urban centres in Rwanda. – Evaluate the impacts of urbanization on development of Rwanda – Analyse the causes and problems of slums in Rwanda and suggest possible solutions	– Demonstrate a continuous commitment to propose solutions to problems caused by urbanization in Rwanda	– Major urban centres e.g.: Kigali, Rubavu, Musanze (location, population, functions) – Factors favouring urbanisation – Problems of the urban centres in Rwanda and the solutions. – Development of slums (examples, causes, problems and possible solutions)	– In group, discuss and compile the finding for a class presentation – Examine and write a report on the impact of urbanisation on the sustainable development of Rwanda. – In groups, use the given photographs and make a discussion on the causes, problems and solutions of the slums in Rwanda. Compile your findings for a classroom presentation
Links to other subjects: <i>Economics: Population studies, Entrepreneurship: Economic activities</i>				
Assessment criteria: <i>The learners are able critically examine the impact of rural and urban settlements on sustainable development in Rwanda</i>				
Teaching and learning materials: <i>Textbooks, internet, illustrations/diagrams, maps, photographs, slide shows, video clips, tactile and talking materials and local environment.</i>				

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S4 Geography		Unit 11: Agriculture in Rwanda		No. of periods: 26
Key unit competence: Examine the impact of various agricultural activities on sustainable development in Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the different types of subsistence cultivation in Rwanda. – Outline the factors favouring the subsistence farming in Rwanda. – Outline the advantages and disadvantages of the subsistence farming in Rwanda. – Identify the major plantation crops in Rwanda, and their characteristics.	– Examine the factors for the subsistence farming in Rwanda. – Evaluate the advantages and disadvantages of the subsistence farming in Rwanda. – Analyse the characteristics of the major plantation crops in Rwanda.	– Appreciate the importance of agriculture to the sustainable development of Rwanda. – Show continual desire to improve the agriculture system in Rwanda.	Crop cultivation Subsistence cultivation: – Meaning and types (small holder farming/ intensive subsistence farming, mixed farming and crop rotation) – Examples of subsistence crops in Rwanda. – Factors for the subsistence farming in Rwanda. – Advantages and disadvantages of intensive subsistence farming	– Use the geographical documents, internet, photographs and maps to analyse different types of cultivation, their characteristics, factors favouring each type of cultivation, advantages and disadvantages of each farming system. In groups, discuss and compile the findings for a class presentation.

– Outline the factors favouring plantation farming in Rwanda. – Outline the advantages and disadvantages of the plantation farming. – Outline the types of crops and characteristics of market gardening and horticulture. – Identify the different types of modern agriculture in Rwanda – Identify the factors for the agricultural modernisation in Rwanda. – Outline the problems limiting the agricultural modernisation in Rwanda.	– Analyse the factors favouring plantation farming in Rwanda. – Evaluate the advantages and disadvantages of plantation. – Describe the types of crops and characteristics of market gardening and horticulture in Rwanda. – Explain the different types of modern agriculture in Rwanda – Explain the factors for agricultural modernisation in Rwanda.	– Show positive attitude toward keeping small livestock in Rwanda	Plantation farming: – Major plantation crops, their characteristics and areas. – Factors favouring plantation agriculture in Rwanda. – Advantages and disadvantages of the plantation farming Case study: – Tea plantation. – Sugar cane plantation (Kabuye/Nyacyonga, Nyabarongo valley). Horticulture and market gardening: – Meaning and characteristics – Types of crops grown under Horticulture and market gardening Modern agriculture in Rwanda:	– Conduct a field study to a plantation farm nearest your school and collect the data on the characteristics, factors favouring the plantation, advantages and disadvantages, the problems related to the plantation farming. Compile the findings for the group discussions and class presentation. – In groups, discuss about the problems limiting agricultural modernisation and suggest the possible ways to develop agriculture in Rwanda. Make a class presentation.
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– Identify the ways of improving agriculture production in Rwanda. – Identify the different systems of rearing animals in Rwanda, and their characteristics. – State the factors hindering the development of dairy farming. – Outline the ways of improving livestock farming in Rwanda. – Identify the types of small animals kept in Rwanda. – State the factors affecting in keeping smaller animals.	– Explain the factors hindering the agriculture modernisation in Rwanda. – Describe the different systems of rearing animals in Rwanda, and their characteristics. – Analyse the factors hindering the development of dairy farming in Rwanda. – Evaluate the factors, methods and the importance of keeping smaller animals in Rwanda.		– Meaning and types of modern agriculture – Factors and methods for the modernisation of agriculture – Problems limiting agriculture modernisation in Rwanda. – Ways of improving agricultural production in Rwanda Livestock farming: – Meaning of ranching, dairying and zero-grazing. – Characteristics of ranching, dairying and zero-grazing – Factors hindering the development of ranching, dairying and zero-grazing – Ways of improving the livestock farming in Rwanda	– Use the geographical documents, internet, photographs and maps analyse different types of livestock farming, their characteristics, factors, problems and ways of improving the livestock farming in Rwanda. In groups, discuss and compile the findings for a class presentation. – By writing an essay, examine the relationship between crop cultivation and livestock farming. – In groups, discuss the contribution of livestock farming to the economy of Rwanda, draw relevant conclusions and make a class presentation.
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– Identify the methods and importance of keeping smaller animals. – Identify the contributions of the livestock farming to the economy of Rwanda. – Identify the problems affecting the livestock farming in Rwanda. – State the importance of agriculture on the sustainable development of Rwanda.	– Analyse the contributions and problems associated with the livestock farming in Rwanda. – Examine the importance of agriculture to the economic development of Rwanda.		Keeping small animals (pigs, poultry, goats, rabbits and apiculture). – Factors affecting the keeping of smaller animals – Methods of improving the keeping of smaller animals. – Importance of keeping smaller animals. – Problems affecting livestock in Rwanda and their solutions – Contributions of the livestock farming to the economy of Rwanda. – The importance of agriculture to the economy of Rwanda.	– In groups, use the knowledge acquired from class, documents, internet, photographs and maps to examine the contribution of crop cultivation and animal rearing to the economy of Rwanda and make a class presentation.
Links to other subjects: <i>Economics: Agricultural production, Mathematics: Statistics.</i>				
Assessment criteria: <i>The learners are able to critically examine the impact of various agricultural activities on sustainable development in Rwanda</i>				
Teaching and learning materials: <i>Geographical documents, maps, tactile and talking materials, concrete objects, photographs, internet, video (audio-visual) clips, local environment and field trip resources.</i>				

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S4 Geography		Unit 12: Forestry in Rwanda	No. of periods:12	
Key unit competence: Evaluate the impact of forestry on sustainable development in Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the term forest and forestry – Identify the major forested areas of Rwanda. – List the factors influencing the forest exploitation. – Outline different methods of lumbering in Rwanda. – Identify the importance of forests and forestry in Rwanda. – List the products of lumbering in Rwanda.	– Describe the characteristics of the major forests in Rwanda. – Explain the factors influencing the forest exploitation. – Describe different methods of lumbering in Rwanda. – Evaluate the importance of forests and forestry in Rwanda.	– Appreciate the importance of conserving forests in Rwanda. – Show concern for the problems affecting forest exploitation and propose mitigation measures.	– Meaning of forest and forestry. – Major forested areas in Rwanda and their characteristics. – Factors influencing the forest exploitation in Rwanda. – Methods of Lumbering in Rwanda – Products from lumbering in Rwanda	– Use the geographical documents, internet, maps, photographs and videos on the major forested areas in Rwanda and describe the factors influencing the forest exploitation, the methods and the importance of lumbering. In groups, discuss the findings for a class presentation.

– Identify the problems affecting forest exploitation in Rwanda. – State the causes and the effects of deforestation in Rwanda. – Identify the measures for forest conservation in Rwanda.	– Analyse the problems affecting the forest exploitation / lumbering in Rwanda. – Examine the causes and the effects of deforestation in Rwanda and suggest the conservation measures.	– Show continual desire to engage in tree-planting campaigns supporting forest conservation and management in Rwanda	– Importance of forests and forestry in Rwanda. – Problems affecting forest exploitation / lumbering in Rwanda. – Deforestation: Causes and effects. – Forest conservation measures in Rwanda in Rwanda	– Use geographical documents, internet, maps, photographs, or videos on the problems affecting forest exploitation, the causes and the effects of deforestation, and the possible control measures for conservation of forests in Rwanda. In groups, discuss the findings, draw relevant conclusions and prepare a written report for a class presentation. – Conduct a field study, observe and record about forest exploitation, and make group discussions. Write a report of the findings for a class presentation.
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Links to other subjects: Biology: *ecology, botany and biomes.*

Assessment criteria: *The learners are able to critically evaluate the impact of forestry on sustainable development in Rwanda*

Teaching and learning materials: *Geographical documents, internet, maps, local environment, tactile and talking materials, field work material and video player.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S4 Geography		Unit 13: Fishing in Rwanda		No. of periods: 8
Key unit competence: Evaluate the impact of fishing on sustainable development in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain concepts of fishing and fish farming – Identify the major fishing grounds in Rwanda. – Identify the factors favouring fishing in Rwanda. – Identify different types of fish and methods of fishing used in Rwanda.	– Locate the major fishing grounds in Rwanda. – Analyse the factors favouring fishing and fish farming in Rwanda. – Describe the methods of fishing in Rwanda. – Evaluate the methods of fish conservation and preservation in Rwanda.	– Appreciate the importance of fish and fishing in Rwanda. – Show concern for the problems affecting fishing and suggest their possible solutions.	– Meaning of fishing – Major fishing grounds of Rwanda – Factors favouring fishing in Rwanda. – Types of fish and the methods of fishing used in Rwanda. – Methods of conservation and preservation of fish in Rwanda. – Importance of fishing in Rwanda.	– Use the geographical documents and internet to identify the major fishing grounds, types of fishes, methods used and factors favouring fishing. In groups, discuss and compile the findings for the class presentation. – In a field study, collect the data on the types of fish, methods of fishing, problems, preservation and conservation methods. In groups, discuss the findings and write a report for a class presentation.

– Outline the methods used for the fish conservation and preservation in Rwanda. – Outline the importance and problems of fishing and fish farming in Rwanda. – Identify different ways of improving fishing and fish farming in Rwanda.	– Explain the importance and problems of fishing and fish farming in Rwanda. – Evaluate the ways and prospects for fishing and fish farming in Rwanda.	– Show continual desire to promote fishing and fish farming in Rwanda.	– Problems affecting fishing and the possible solutions. Fish farming: – Meaning and factors favouring fish farming in Rwanda. – Problems and prospects for fish farming in Rwanda. Case study: fishing on lake Kivu	– In group work, discuss the importance and the problems of fish and fishing on the economy of Rwanda. Make a report for a class presentation. – Use the geographical documents and internet to examine the importance, factors favouring fish farming, problems affecting fish farming and prospects of fish farming in Rwanda. In groups, discuss the findings and make a class presentation. – In groups, suggest recommendations that can improve fishing in Rwanda and write a report.
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Links to other subjects: *Biology: Zoology (aquatic life)*

Assessment criteria: *The learners are able to critically evaluate the impact of fishing on sustainable development in Rwanda*

Teaching and learning materials: *Documents, internet, photographs, local environment, fieldwork materials, brailled textbooks, fishing materials.*

Topic Area: Human and Economic Geography			Sub-Topic Area: Economic activities	
S4 Geography		Unit 14: Mining in Rwanda		No. of periods: 7
Key unit competence: Examine the impact of mining on sustainable development in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the major types of minerals mined in Rwanda – State the methods used in mining in Rwanda. – Outline the factors favouring mineral exploitation in Rwanda. – Identify the importance of mining to the economy of Rwanda.	– Locate the major minerals and their mining sites on a sketch map of Rwanda. – Analyse the methods used in the mining in Rwanda. – Examine factors favouring mineral exploitation of in Rwanda. – Evaluate the importance of mining to the economy of Rwanda.	– Appreciate the importance of minerals. – Participate in raising community awareness on the importance of environmental protection around mining sites – Show respect for the methods used to restore the damaged environment.	– Types and distribution of the major minerals in Rwanda. – Methods of mining in Rwanda. – Factors favoring the exploitation of minerals in Rwanda – Importance of mining to the economy of Rwanda.	– Use geographical documents, internet, mining map of Rwanda and photographs to identify the major mining areas, methods and factors affecting mining in Rwanda. In groups, discuss and compile the findings for a class presentation. – Use the geographical documents, internet, map of Rwanda, photographs or field study observation examine the importance and problems affecting mineral exploitation, and their possible solutions. In groups, discuss and compile the findings, and write an essay.

– Identify the problems affecting the mining and their possible solutions. – Outline the effects of mining on the environment and their mitigation measures	– Analyse the problems affecting the mining in Rwanda and suggest their possible solutions. – Examine effects of mining on the environment and propose their mitigation measures		– Products from the minerals in Rwanda. – Problems affecting the mining and their possible solutions. – Effects of mining on the environment and their mitigation measures	– In pairs, discuss whether it is important for the government to regulate and control the mineral exploitation in Rwanda. Write a report.
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Links to other subjects: *Chemistry: Transitional metals, mineral properties.*

Assessment criteria: *The learners are able to critically examine the impact of mining on sustainable development in Rwanda*

Materials: *Textbooks/ brailled textbooks, journals, photographs, local environment, maps, tactile and talking materials, internet/Jaws software and field study resources.*

Topic Area: Human and Economic Geography			Sub-Topic Area: Economic activities	
S4 Geography			Unit 15: Power and Energy in Rwanda	No. of periods: 10
Key unit competence: Examine the impact of power and energy production on sustainable development of Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify with examples the major sources energy (the renewable and non-renewable) used in Rwanda – Outline the factors favouring the power production in Rwanda.	– Distinguish between the renewable and non-renewable sources of energy in Rwanda. – Analyse the factors favouring power production in Rwanda.	– Show a continual desire to advocate for the use of renewable energy sources that are friendly to the environment – Appreciate the importance of power and energy on sustainable development of Rwanda.	– Major sources of energy used in Rwanda (renewable and non-renewable) and their examples. – Factors favouring the power production in Rwanda. – Importance of power in the development of Rwanda.	– Use geographical documents, internet, map of Rwanda and photographs to identify the major sources of energy used in Rwanda, the factors favouring the power production and the importance of energy. In groups, discuss and compile the findings for a class presentation. – Use internet and textbooks to examine the problems hindering the production of power and suggest possible solutions. In groups, discuss and compile the findings for a class presentation.

– Identify the importance of power and energy in Rwanda. – Identify the problems hindering the power production and their possible solutions in Rwanda	– Evaluate the importance of power and energy in Rwanda. – Investigate the problems hindering the development of energy and suggest possible solutions.	– Show commitment to support the development of power and energy in Rwanda	– Problems hindering the power production and their possible solutions in Rwanda. – Case study: Mukungwa I, Kivu Watt and Rwamagana Solar Power Station	– Conduct a field study at any power plant, observe and record the findings about the power and energy production. Make a report for a class presentation. – In pairs, analyse and write a report on how Rwanda cope with the increasing demand for power with an increasing population will?
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Links to other subjects: *Physics: Electricity, power and energy.*

Assessment criteria: *The learners are able to critically examine the impact of power and energy production on sustainable development of Rwanda*

Teaching and learning materials: *Textbooks/ brailled textbooks, journals, photographs, maps, tactile and talking materials, internet/Jaws software and field study resources.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S4 Geography		Unit 16: Industry in Rwanda		No. of periods: 12
Key unit competence: Analyze the impact of industrialization on sustainable development in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the concepts of industry and industrialisation. – List the types of industries in Rwanda. – Outline the factors influencing the location of industries in Rwanda. – Identify the importance of industries in Rwanda.	– Categorize the types of industries and their industrial products made in Rwanda. – Examine the factors influencing the location of industries in Rwanda. – Evaluate the importance of industries in the economic development of Rwanda.	– Appreciate the importance of industrial development to the economic development of Rwanda – Show interest in addressing the challenges affecting industrial growth in Rwanda	– Meaning of industry and industrialisation. – Types of industries and their industrial products in Rwanda. – Factors influencing the location of industries and industrial development in Rwanda. – Importance of the industries in Rwanda.	– Use the geographical documents, internet, map of Rwanda, and photographs to analyse the types of industries, factors for the location, factors influencing the industrial development and the importance of industries in Rwanda. In groups, discuss and compile the findings for a class presentation.

– Identify the problems affecting the industrial development in Rwanda. – Outline the problems associated with industrialisation in Rwanda and the ways to mitigate them.	– Examine the problems affecting the industrial development in Rwanda. – Analyse the problems associated with industrialisation in Rwanda and the ways to mitigate them.	– Show continual desire to respond on the challenges of industrialization in Rwanda by proposing possible measures to address them	– Problems affecting the industrial development and their possible solutions in Rwanda. – Problems associated with industrialisation in Rwanda and the ways to mitigate them. Case study: – Mulindi Tea Factory – BRALIRWA – CIMERWA	– In groups, discuss the problems associated with industrialisation in Rwanda and the ways to mitigate them. Compile the findings for a class presentation. – Conduct a field study of an industry in the local area, observe and record findings related to industrialization for class discussion, and write a report with relevant conclusions to present to the class
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Links to other subjects: *Economics: Industrial production in Economics.*

Assessment criteria: *The learners are able to critically analyse the impact of industrialization on sustainable development in Rwanda.*

Teaching and learning materials: *Textbooks/ brailled textbooks, journals, photographs, maps, local environment, tactile and talking materials, internet/Jaws software and field study resources.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S4 Geography		Unit 17: Transport, Communication and Trade in Rwanda		No. of periods: 12
Key unit competence: Examine the impact of transport, communication, and trade on sustainable development in Rwanda				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– State different types of transport in Rwanda. – Outline the advantages and disadvantages of the different types of transport in Rwanda. – Identify the factors influencing the development of the transport.	– Locate the international airports, national roads, and major waterways used for transport in Rwanda – Analyse the factors influencing the development of transport in Rwanda.	– Appreciate the importance of promoting the development of transport in Rwanda. – Appreciate the importance of communication in Rwanda.	Transport – Major types of transport in Rwanda. – Advantages and disadvantages of each type of transport. – Factors influencing the development of transport. – Importance of transport on the sustainable development of Rwanda.	– Use the geographical documents, internet, map of Rwanda and photographs to analyse the major types of transport, factors influencing transport, importance of transport, problems and the possible solutions. In groups, discuss and compile the findings for a class presentation.

– Identify the importance of transport in Rwanda. – Identify the problems affecting the transport in Rwanda. – List different means of communication used in Rwanda. – Identify the factors influencing the development of communication in Rwanda, – Identify the importance of communication in Rwanda. – Outline the problems affecting communication and the possible solutions.	– Evaluate the advantages and disadvantages of the different types of transport in Rwanda – Explain the importance of transport in Rwanda – Examine the problems of transport and the prospects. – Differentiate the means of communication used in Rwanda. – Explain the factors influencing the development of communication. – Evaluate the importance of communication in Rwanda.	– Show continual desire to use different means of communication for enhancing the development of Rwanda. – Show continual desire to acquire entrepreneurial skills in trade for the development in Rwanda.	– Problems affecting the transport (land locked position, rugged relief, shortage of capital). – Possible solutions and the prospects of transport in Rwanda. Communications – Different means of communication. – Factors influencing the development of communication in Rwanda – Importance of communication in Rwanda – Problems affecting communication and their possible solutions	– In groups, discuss about the problems affecting the transport and debate possible solutions and the prospects of transport in Rwanda. compile the findings for a class presentation – Use the geographical documents, internet, map of Rwanda and photographs to describe different means of communication, factors influencing the development of communication, importance of communication, problems and possible solutions. In groups, discuss and compile the findings for the class presentation.
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– Explain the meaning of internal and external trade, importation and exportation – Identify the factors affecting trade in Rwanda. – Identify the importance of trade in Rwanda – Explain the balance of trade and the balance of payment in Rwanda. – Identify the problems affecting the trade in Rwanda, their possible solutions and prospects	– Analyse the problems and solutions of communication. – Analyse the factors affecting trade in Rwanda. – Evaluate the importance of trade in Rwanda. – Analyze Rwanda's balance of trade and balance of payments. – Describe the problems affecting trade in Rwanda and suggest possible solutions		Trade – Internal and external trade in Rwanda. – Importation and exportation of the products (balance of trade and balance of payment). – Factors affecting trade in Rwanda. – Importance of trade in Rwanda. – Problems affecting trade in Rwanda, possible solutions and prospects.	– Use documents, internet, map of Rwanda and photographs to identify the factors affecting trade, imports and exports, importance and problems of trade. In groups, discuss and compile the findings for a class presentation. – In pairs, discuss and write a report on the impact transport, communication and trade on the sustainable development of Rwanda.
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Links to other subjects: *Economics: International trade, Mathematics: statistics*

Assessment criteria: *The learners are able to properly examine the impact of transport, communication, and trade on sustainable development in Rwanda*

Teaching and learning materials: *Textbooks/ brailled textbooks, photographs, local environment, maps, tactile and talking materials, internet/Jaws software.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic Activities		
S4 Geography		Unit 18: Environmental Conservation in Rwanda and Tourism		No. of periods: 10
Key unit competence: Evaluate the impact of environmental conservation and tourism on sustainable development in Rwanda.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the concepts of environmental conservation, tourism and eco-tourism – List different natural resources of Rwanda. – Identify the reasons and ways for the conservation of environment in Rwanda.	– Classify the different natural resources of Rwanda. – Evalaute the reasons and ways for the environment conservation. – Examine the problems affecting the conservation of environment and propose ways of conservating natural resources.	– Show desire to protect and conserve various tourist attractions available in Rwanda. – Appreciate the importance of environmental conservation in the economy of Rwanda.	Environmental Conservation – Meaning of environmental conservation. – Types of natural resources (renewable and non-renewable). – Reasons for the conservation of natural resources in Rwanda.	– Use the geographical documents, internet, map of Rwanda, photographs or data from field to identify the reasons and ways for the conserving the environment, impact of environmental conservation and the problems encountered. In groups, discuss and compile the findings for the class presentation.

– Identify the factors and the problems affecting the conservation in Rwanda. – outline the major tourist attractions in Rwanda. – Identify the factors affecting tourism in Rwanda. – Identify the importance of tourism in Rwanda. – Identify the problems of tourism in Rwanda.	– Locate the major tourist attractions in Rwanda. – Explain the factors affecting the development of tourism in Rwanda. – Evaluate the impact of tourism on the environment and development in Rwanda. – Explain the problems affecting tourism and suggest the possible solutions. – Justify the importance of tourism in the development of the economy in Rwanda.	– Show commitment to address environmental problems and advocate for the sustainable protection of the environment – Appreciate the importance of tourism on the sustainable development of Rwanda. – Show concern for the problems affecting tourism and suggest possible solutions	– Ways of conservating natural resources (wetlands, forests, wildlife, land and minerals, water resources, aquatic animals) in Rwanda. – Problems encountered in conserving the environment and their solutions in Rwanda. Tourism – Meaning of tourism, eco-tourism and the forms of tourism. – Major tourist attractions in Rwanda. – Factors affecting the development of tourism in Rwanda. – Importance and prospects of tourism in Rwanda.	– Use the geographical documents, internet, map of Rwanda, photographs or data from the field study and identify the major tourist attractions, the factors affecting the development of tourism, the importance and the problems affecting the tourism. In groups, discuss and compile the findings for a class presentation. – Evaluate the relationship between the environmental conservation and tourism on the sustainable development of Rwanda.
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– Outline the importance of tourism on the environment and economy of Rwanda.			– Problems affecting the tourism in Rwanda and the solutions. – Impact of tourism on the environment and the development in Rwanda Case study: – Nyungwe National Park – Akagera National Park – Birunga National Park	In pairs, write a report. – Conduct a field study to a local tourist site and observe and record the information about tourism. Draw relevant conclusions in a report.
Links to other subjects: Biology: <i>Ecology</i>				
Assessment criteria: <i>The learners are able to properly evaluate the impact of environmental conservation and tourism on sustainable development in Rwanda.</i>				
Teaching and learning Materials: <i>Textbooks/ brailled textbooks, photographs, local environment, maps, tactile and talking materials, internet/Jaws software.</i>				

SENIOR FIVE

6.3. GEOGRAPHY FOR SENIOR 5

6.3.1. General competences of geography at the end of senior five:

- Explain different physical geographical phenomena and their impact on the sustainable development in the world.
- Explain different problems associated with the physical, human and economic environment of the world and suggest solutions.
- Evaluate the major challenges related to the population growth and exploitation of the resources in the world.
- Compare different modes of development in the world.
- Use statistical diagrams and maps to interpret the geographical information.

6.3.2. Syllabus units for senior five

Unit	Unit title	Periods
1	<i>Statistical graphs, charts and maps in geography</i>	22
2	<i>Map reading and interpretation</i>	19
3	<i>The universe and the solar system</i>	10
4	<i>Origin and structure of the earth</i>	7
5	<i>Internal landform processes</i>	32
6	<i>Soils</i>	7
7	<i>Weather and climate of the world</i>	21
8	Natural vegetation of the world	10
9	Population growth in the world	16
10	Rural settlement and urbanization in the world	14
11	Agriculture in the world	24
12	Forestry in the world	16
13	Fishing in the world	17
14	Mining in the world	16

GEOGRAPHY SYLLABUS FOR SENIOR FIVE

Topic Area: Practical Geography		Sub-Topic Area: Field work and statistics		
S5 Geography		Unit 1: Statistical graphs, charts and maps in Geography		No. of periods: 16
Key unit competence: Interpret geographical data using statistical graphs, charts and maps				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the importance of statistics in geography. – Outline the different types of statistical graphs, charts and maps used in geography.	– Construct the statistical graphs and charts and maps to present the various geographical data. – Explain the advantages and disadvantages of various statistical graphs, charts and maps.	– Appreciate the importance of statistics in interpreting and solving geographical problems.	– Importance of statistics in Geography. Line graphs: – Simple line graph – Advantages and disadvantages of simple line graphs – Group line graphs – Advantages and disadvantages of group line graphs – Divergence line graphs	– Use statistical and geography textbooks to identify advantages and disadvantages of the statistical graphs, charts and maps. In groups, discuss the findings and make a class presentation.

– Identify the advantages and disadvantages of each statistical graph, chart and map.	– Draw different statistical graphs, charts and maps. – Interpret different geographical data using statistical graphs, charts, and maps.		– Advantages and disadvantages of divergence line graphs Bar graphs – Simple bar graph – Advantages and disadvantages of simple bar graphs – Group / comparative bar graph – Advantages and disadvantages of group bar graphs – Age and sex graphs age and sex graphs – Advantages and disadvantages of age and sex graphs – Divergence bar graphs – Advantages and disadvantages of divergence bar graphs Statistical charts: – Divided circles/ Pie charts	– Use the statistical data (geographical data) to construct and interpret different graphs (line and bar graphs), charts (divided circles/pie-chart) and maps (dot, isoline and shading/ choropleth maps). In groups, make a class demonstration on how to construct different graphs, charts and maps – In pairs, discuss the advantages and disadvantages of each statistical graph, chart and map. Write down the findings and make a class presentation.
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			– Advantages and disadvantages of divided circles/ Pie charts Statistical maps: – Dot maps – Isoline maps – Shading or choropleth maps – Advantages and disadvantages of Statistical maps.	
Links to other subjects: <i>Mathematics: Descriptive statistics.</i>				
Assessment criteria: <i>The learners are able to correctly interpret geographical data using statistical graphs, charts and maps</i>				
Teaching and learning materials: <i>Geographical documents, maps, journals, tactile and talking materials, internet, statistical data, diagrams and graph papers.</i>				

Topic Area: Practical Geography		Sub-topic Area: Map reading and photograph interpretation		
S5 Geography		Unit 2: Map reading and interpretation		No. of periods: 28
Key unit competence: Determine the bearings, directions, distances, areas, and the relationships between physical features and human activities on a topographic map				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify Eastings and Northings on a topographic map. – Identify the directions and bearings on the topographic maps.	– Locate the places using the geographical coordinates/grid reference systems (northings and Eastings). – Determine bearings and directions on topographic maps – Measure and calculate the distances and surface areas on topographic maps.	– Show interest in using topographic maps to interpret geographical phenomena and the interaction between relief and human activities. – Appreciate the importance of demonstrate the relationship between real-world distances and map distance	– Location of places using grid references. – Directions and bearings on a topographic map – Measuring distances on maps (straight line, curved line – Calculating areas on a topographic map (regular and irregular shapes). – Representation of relief on the map (contours, colouring and shading)	– Use Northings and Eastings (grid references) to locate places on the maps. – Use mathematical set and drawing equipment to identify and connect points or places on a topographic map. – Use mathematical equipment like pair of compasses, rulers, protractors and dividers to calculate the directions and bearings on the topographic maps.

– Identify the physical and human features on topographic maps using contours, spot heights, trigonometrical stations and the colours.	– Examine contours, coloring, and shading on maps to determine the types of relief features represented. – Determine the vertical intervals and amplitude of relief /spot heights. – Interpret the relationship between physical features and human features/ aspects on a topographic map	– Demonstrate a positive attitude toward using geographical coordinates and a compass for accurate location on topographic maps	– Calculation of vertical intervals and amplitude of relief /spot heights. – Interpretation of physical aspects from maps – Interpretation of human aspects on maps – Relationship between physical and human aspects on maps	– Use a topographic map, key, the scale and the mathematical set to identify the areas, lines, roads or shapes on the maps, and measure and calculate distance and areas on a map. – Use a calculating tool/ calculator to calculate the vertical intervals and amplitude of relief/ spot heights. – In groups, discuss the findings and make a class demonstration
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Links to other subjects: *Mathematics: Trigonometry, points, lines, distances and areas*

Assessment criteria: *The learners are able to measure and determine clearly the bearings, directions, distances, areas, and the relationships between physical features and human activities on a topographic map.*

Teaching and learning materials: *Geographical documents, journals, internet, tactile and talking materials, topographic maps, tactile maps, mathematical instruments and graph papers.*

Topic Area: Physical Geography		Sub-topic Area: Understanding the earth and universe		
S5 Geography		Unit 3: The Universe and the Solar System		No. of periods: 7
Key unit competence: Analyze the components of the universe, solar system and the consequences of the Earth’s movements.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the components of the universe and the solar system. – Identify the planets found in the solar system. – Explain the peculiar elements of the Earth. – Outline the characteristics of each planet of the solar system.	– Describe components of the universe (constellations and Galaxies). – Explain the influence of the sun on the earth. – Describe the characteristics of the different planets and the peculiar elements of the earth.	– Appreciate the need for the sustainable practices to protect the Earth’s environment. – Show a curiosity to discover more objects that exist in the universe.	– Meaning and components of the universe (constellations and Galaxies) – The solar system (sun and planets) – Peculiar elements of the earth – Earth’s movements (rotation and revolution) and their consequences	– Using internet, textbooks, satellite photos and the diagrams or power point presentation/slides to analyse the components of the universe (Cluster, galaxies, constellations/stars, planets, satellites, Asteroids, comets, meteors and Meteorites) and their characteristics. In groups, discuss and compile the findings for a class presentation. – In pairs, use internet and the geographical documents to discuss about the movements of the earth and their related consequences. Compile the findings for a class presentation.

– Identify the consequences of the earth's movements. – State the characteristics, phases and effects of the moon on Earth. – Identify other heavenly bodies.	– Examine the consequences of the earth's movements. – Describe other celestial/ heavenly bodies in the universe (Asteroids, comets, meteors and Meteorites)	– Demonstrate appreciation for the differences and uniqueness of the Earth compared to other planets.	– The moon (characteristics, phases and influence of the moon on Earth) – Other heavenly bodies: Asteroids, comets, meteors and Meteorites	– Use the geographical documents, internet, atlases and satellite photos to analyse the characteristics, movements, phases and influence of the moon on the earth. In groups, discuss the findings and make a class presentation. – Discuss in groups, the characteristics of other heavenly bodies (Asteroids, comets, meteors and Meteorites). Present the findings to the class.
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Links to other subjects: *Physics: distances, gravity, velocity and volumes*

Assessment criteria: *The learners are able to analyze critically the components of the universe, solar system and the consequences of the Earth's movements*

Teaching and learning materials: *Geographical documents, maps, atlas, journals, tactile and talking materials, journals, aerial photographs and internet/Jaws software.*

Topic Area: Physical Geography		Sub-topic Area: Understanding the earth and universe		
S5 Geography		Unit 4: The Origin and structure of the Earth		No. of periods: 7
Key unit competence: Explain the main theories of the origin of the Earth and its structure				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify main theories of the origin of the earth (Big bang and creation, theory/biblical theories). – Identify the Characteristics of the Earth. – Identify the components of the internal structure of the Erath.	– Explain how the Earth came into existence using different scientific theories – Describe the characteristics of the (Shape and evidence to prove that the earth is spherical, size, diameter, circumference, volume, mass and gravity) Earth	– Appreciate the importance of the earth as the only planet in the solar system that supports life. – Show continual desire to protect the Earth (the only home for flora and fauna).	– Theories of the origin of the earth (Big bang and creation/ biblical theories). – Characteristics (Shape and evidence to prove that the earth is spherical, size, diameter, circumference, volume, mass and gravity) of the Earth.	– In groups, use the geographical documents, internet, the globe and the satellite photos to explain the theories explaining the origin of the earth, shape of the earth and evidence to prove that the earth is spherical. Compile the findings and present them to the class – In pairs, use the internet and textbooks to explain more about the earth's internal structure, mineral composition and the superficial configuration of the earth. In groups, discuss and compile the above findings for a class presentation.

– State the major geological eras and the periods.	– Illustrate the internal structure, mineral composition and the superficial configuration of the earth. – Explain the geological time.		– The internal structure of the earth. – Geological time scale	– In groups, use the geological tables, internet and textbooks to find out the geological eras and the periods. Discuss, write down the findings and make a class presentation.
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Links to other subjects: *Mathematics: Calculating diameter, circumference, volume, mass and gravity of spheroids /spheres.*

Assessment criteria: *The learners are able to properly explain the main theories of the origin of the Earth and its structure*

Teaching and learning materials: *Geographical materials, maps, tactile and talking materials, globes, atlases, journals, photographs, internet/Jaws software and local environment.*

Topic Area: Physical Geography		Sub-topic Area: Landform evolution and processes		
S5 Geography		Unit 5: Internal landform Processes		No. of periods: 32
Key unit Competence: Analyse the internal processes responsible for the formation and evolution of various relief landforms				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the processes leading to the formation of different faulted features. – Identify the major types of faults and landforms associated with faulting. – Identify the effects of faulted landforms on human activities.	– Describe the processes responsible for the formation of different faulted landforms. – Evaluate the impact of faulting on the landscape and drainage. – Analyse the effects of faulted landforms on human activities. – Locate the different areas associated with faulting in the world.	– Show concern to protect the landforms resulted from faulting processes – Show appreciation for the importance of faulting processes on landscapes and drainage on human activities	Faulting – Processes/ factors leading to the formation of different faulted features – Types of faults – Influence of faulting on landscape and drainage – Distribution of landforms associated with faulting – Impact of faulting on human activities	– Use different geographical documents (textbooks) and internet to explain on the processes of faulting, types of faults and the major faulted landforms. In group work, discuss the findings and make a class presentation. – In pairs, use textbooks and internet to analyse the impacts of faulting on drainage and human life. Discuss the findings and make a class presentation. – In groups, use the map of the world and locate main faulted landforms. Compile the findings and make a presentation to the class.

– Explain the processes of the folding, types of folds and its influence on the drainage. – Identify the major landforms and drainage features associated with the folding.	– Examine the processes responsible for the folding. – Describe the landforms and drainage features associated with folding.	– Show concern to protect the landforms resulted from folding processes – Show appreciation for the importance of folding processes on landscapes and drainage on human activities	Folding – Processes leading to the formation of different folded features – Types of folds – Influence of folding on landscape and drainage – Distribution of landforms associated with folding	– Use different geographical documents (textbooks) and internet to explain on the processes of folding, types of folds and the major folded landforms. In group work, discuss the findings and make a class presentation. – In pairs, use textbooks and internet to analyse the impacts of folding on drainage and human life. Discuss the findings and make a class presentation. – In groups, use the map of the world and locate main folded landforms. Compile the findings and make a presentation to class
– Explain the term warping and its associated landforms.	– Analyse Causes/factors leading to the occurrence of upwarping and down warping processes	– Show concern to protect the landforms resulted from warping processes	Warping – Causes/factors leading to the occurrence of upwarping and down warping processes	– Use different geographical documents (textbooks) and internet to explain the processes of warping (up and down warping) and major landforms associated to warping. In group work, discuss the findings and make a class presentation.

– Identify the landforms and drainage features associated with process of upwarping and down warping.	– Describe landforms and drainage features associated with upwarping and down warping	– Show appreciation for the importance of warping processes on landscapes and drainage on human activities	– Upwarping and down warping associated landforms (plateaus, geosyncline, basins and depression/basin lakes)	– In pairs, use textbooks and internet to analyse the impacts of warping on drainage and human life. Discuss the findings and make a class presentation.
– Explain the concept of vulcanicity, vulcanicity and volcanic processes. – Identify the major landforms associated with vulcanicity and volcanicity. – Outline the types of volcanoes and their characteristics.	– Describe the different relief landforms associated with the intrusive and extrusive processes. – Describe the types and characteristics of different volcanoes. – Explain the impact of different volcanic landforms on human life.		Vulcanicity and Volcanicity: – Processes of vulcanicity and volcanic materials – Intrusive features – Volcanic/ extrusive features – Types of volcanoes and their characteristics	– Use different geographical documents (textbooks) and internet to explain on the processes of vulcanicity/ vulcanicity, types of volcanoes and intrusive and extrusive volcanic landforms. In group work, discuss the findings and make a class presentation. – In pairs, use textbooks and internet to analyse the impacts of vulcanicity on drainage and human life. Discuss the findings and make a class presentation.

– Explain the impact of volcanicity on the drainage and to the human life. – Identify the major volcanic regions in the world.	– Locate on the world map the major volcanic regions of the world.		– World distribution of volcanoes and other volcanic features – Impact of volcanicity on environment and human activities	– In groups, use the map of the world and locate main volcanic landforms. Compile the findings and make a presentation to the class.
– Explain the meaning of earthquakes. – Outline the causes and consequences of earthquakes. – Identify the measures to curb the earthquake effects. – Identify the major earthquake zones of the world.	– Evaluate the causes and the effects of the earthquakes. – Propose possible measures to regulate earthquake effects. – Locate the map of the world the major earthquake zones of the world.	– Show concern to participate in reducing consequences of earthquake impacts and promote preparedness.	Earthquake: – Process of earthquake and its related concepts – Measurement/ recording of earthquake magnitude and intensity – Causes and consequences of earthquakes – Measures/ solutions to control the consequences of earthquakes	– Use different geographical documents (textbooks) and internet to explain the process of earthquakes, its causes and consequences of earthquakes. In group work, discuss the findings and make a class presentation. – In pairs, use textbooks and internet to analyse measurement/recording of earthquake magnitude and intensity, and measures/ solutions to control the consequences of earthquakes. Discuss the findings and make a class presentation.

			– World distribution of earthquakes	– Use the map of the World and locate areas that are mostly affected by earthquakes/tsunami
Links to other subjects: <i>Physics: Forces (compressional and tensional forces).</i>				
Assessment criteria: <i>The learners are able to critically analyse the internal processes responsible for the formation and evolution of various relief landforms.</i>				
Teaching and learning materials: <i>Geographical documents, photographs, maps, journals, atlas, local environment, charts, internet website /jaws software, tactile and talking materials.</i>				

Topic Area: Physical Geography		Sub-topic Area: Soils		
S5 Geography		Unit 6: Soils		No. of periods: 7
Key unit competence: Analyze the different constituents and morphological properties of soil.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the constituents of the soil. – Outline the different morphological properties of the soil. – Explain soil profile, soil horizons, soil catena, and soil fertility.	– Examine the main constituents of soil – Describe the properties of the soil. – Describe how soil changes along a slope (catena).	– Appreciate the importance of soil constituents in agriculture – Show continual desire to willingly participate in protecting and conserving soil.	– Soil constituents – Morphological/ physical and chemical properties of soil (pH and soil nutrients) – Soil profile, catena and fertility	– Use geographical textbooks and internet to identify and explain the constituents of the soil and chemical and morphological properties of the soil. In groups, discuss the findings for a class presentation. – In groups, discuss the concepts of soil profile, soil structure, soil texture, soil porosity, soil pH, soil catena and the soil fertility. Write down the findings and make a class presentation.
Links to other subjects: Chemistry: <i>Acids and bases (pH: potential of hydrogen)</i>				
Assessment criteria: <i>The learners are able to critically analyze the different constituents and morphological properties of soil.</i>				
Teaching and learning materials: <i>Documentaries, journals, soil maps, soil samples, photographs, atlas, internet, local environment, tactile and talking materials.</i>				

Topic Area: Physical Geography		Sub-topic Area: Weather and climate		
S5 Geography		Unit 7: Weather and Climate of the world		No. of periods: 21
Key unit Competence: Analyze the impact of climate on the environment and human activities across the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of the atmosphere. – Identify the layers and composition of the atmosphere. – Identify the gases that compose the Earth’s atmosphere – Identify the importance of the atmosphere.	– Illustrate the structure of atmosphere – Describe the composition of the atmosphere – Examine the importance of atmosphere – Measure and record temperature, humidity, precipitation, wind direction, wind speed, cloud cover, sunshine and atmospheric pressure.	– Appreciate the importance of atmosphere on humana life – Show continual desire to protect the atmosphere/ environment. – Appreciate the importance of the weather elements on the human activities.	– Meaning and structure of the atmosphere – The composition of the atmosphere – Importance of atmosphere – Elements of weather and climate – The measuring /recording instruments of weather elements	– In groups, use the geographical documents, internet photos and diagrams to illustrate the structure of the atmosphere and describe its composition and its importance. Write down the findings and make a class presentation. – Use the geographical documents, textbooks, internet, maps and photographs on the elements of weather and their measurements to identify the instruments used to measure weather elements. In groups, use discuss the findings and make a class presentation.

– Explain the meaning of the concepts of weather and climate – Explain the elements of weather and their measuring/ recording instruments. – Identify the factors that influence each weather element/ condition/ parameter – Meaning and effects of air masses on weather and climate	– Analyse the factors that influence temperature, humidity, precipitation, wind direction, wind speed, cloud cover, sunshine and atmospheric pressure of given area – Analyse the effects of cyclones and anticyclones on weather and environment – Describe the major types of air masses – Analyse the effects of air masses on weather and climate	– Show a continual desire to obtain updated weather information from meteorological agencies and to utilize weather data in everyday life. – Show concern for the causes and the effects of climate change and support to preserve or retain the stability of the atmosphere.	– Factors that influence each weather element/ condition/ parameter – Meaning and effects of air masses on weather and climate – Types of world climates/ climatic zones and their characteristics – Factors that influence World Climates – The relationship between climate and human activities	– Visit the nearest weather station then observe the available instruments and how they are used to measure and record weather elements (temperature, humidity, precipitation, wind direction, wind speed, cloud cover, sunshine and atmospheric pressure of given area). In groups, record information from the weather station and compile findings for class presentation. – In groups, use geographical documents and internet to explain the types of winds, the factors influencing nature and the movement of the winds. Write down the findings for a class presentation. – In pairs, use textbooks and internet to describe the types, characteristics and effects of air masses, cyclones and anti-cyclones on weather and human activities. In groups, discuss the findings and make a class presentation.
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- Name the major climatic zones of the world and outline the characteristics of each zone. - List the factors that influence the world climates. - Outline the effects of climate on human activities.	- Locate on the map of the world the types of world climates/climatic zones - Explain characteristics of tropical, temperate and cold climatic zones/climates - Explain the factors that influence World Climates - Examine the relationship between climate and human activities			- Use the geographical documents to explain the types of humidity, factors influencing humidity and its significance on the environment. In groups, discuss and compile the findings and make a class presentation. - In groups, use internet and geographical documents to explain the types of clouds, factors influencing cloud formation and their effects on the weather. In groups, discuss and compile the above findings and make a class presentation - Use the geographical documents and internet to discuss on the atmospheric pressure, location of the major pressure zones and their characteristics. In groups, discuss and compile the findings and make a class presentation.
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				– Use internet and the geographical documents to describe the major climatic zones of the world, their characteristics, factors influencing the climate and the influence of climate on the human activities. In groups, discuss and compile the above findings and make class presentation. – In pairs, discuss whether the world climates/climatic zones affect human activities across the world. Record your findings in a report and submit it.
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Links to other subjects: *Mathematics: Statistics (sum, average and mean), Physics: pressure, velocity, heat, temperature and energy transfer (conduction, convection and radiation)*

Assessment criteria: *The learners are able to critically analyze the impact of climate on the environment and human activities across the world.*

Teaching and learning materials: *Geographical documents, internet, maps, local environment, instruments (thermometer, rain gauge, barometer), atlas, journals, tactile and talking materials, diagrams and photographs.*

Topic Area: Physical Geography		Sub-topic Area: Vegetation		
S5 Geography		Unit 8: Natural Vegetation of the world		No. of periods: 10
Key unit Competence: Analyze the distribution, characteristics and importance of different types of vegetation in the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– State the location of major type of natural vegetation of the world – Outline the characteristics of each type of natural vegetation (tropical, temperate, grasslands, temperate grasslands, desert, mountain and aquatic vegetation).	– Locate on the map of Africa the major types of major types/ zones of world natural vegetation. – Describe the characteristics of each type of natural vegetation (tropical, temperate, grasslands, temperate grasslands, desert, mountain and aquatic vegetation).	– Appreciate the importance of the various types of vegetation in the world. – Show a continual desire to participate in activities of conservation and management of natural vegetation.	– Tropical rainforests: Characteristics and location – Temperate forests: Characteristics and location – Tropical grasslands: Characteristics and location – Temperate grasslands: Characteristics and location – Desert vegetation: Characteristics and location – Mountain vegetation: Characteristics and location	– Use the geographical documents, internet, maps, power point / slide presentation and photographs to describe the types of vegetation, location and their characteristics. In groups, discuss and write down the findings and make a class presentation. – Use documents, textbooks, internet, maps and photographs to discuss on the factors influencing the vegetation distribution and the importance of vegetation. In groups, discuss the findings and make a class presentation.

– Identify the factors influencing the vegetation distribution. – Outline the importance of the natural vegetation.	– Explain the factors influencing the vegetation distribution. – Analyse the importance of the natural vegetation.		– Aquatic/Marsh vegetation: Characteristics and location – Factors influencing the vegetation distribution – Importance of the natural vegetation	– In pairs, discuss how bad is deforestation and logging activities affect human activities and the environment. Record the findings as report and make a class presentation.
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Links to other subjects: *Biology: Ecology and Botany (plant species).*

Assessment criteria: *The learners are able to critically analyse the distribution, characteristics and importance of different types of vegetation in the world*

Teaching and learning materials: *Geographical documents, internet websites, journals, maps, photographs, atlas, tactile and talking materials, documentary films, diagrams and local environment.*

Topic Area: Human and Economic Geography		Sub-topic Area: Population		
S5 Geography		Unit 9: Population Growth in the world		No. of periods: 16
Key unit competence: Analyze the problems associated with population growth and the strategies of controlling it globally				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of human diversity, optimum population, under population, over population and population density. – Identify the factors that influence the population distribution in the world.	– Describe human diversities of the world – Analyse the world population distribution and density – Explain the meaning of optimum population, under population and overpopulation	– Appreciate the importance of human diversity in development. – Show respect to the various policies for controlling the population growth.	– Human diversities – World population distribution and density – Meaning of optimum population, under population, overpopulation – Effects of underpopulation and overpopulation.	– Use internet, geographical documents, population maps and photographs to explain the population concepts (diversity, density, optimum population, over population and underpopulation). Write down the findings and make a class presentation – In pairs, discuss the factors influencing world population distribution. In groups, discuss the findings and make a class presentation

– Identify the effects of underpopulation and overpopulation – Outline the population problems of the developing and developed countries. – Outline the factors for a rapid population growth – List causes, effects and possible strategies of population growth	– Examine the effects of underpopulation and overpopulation. – Analyse causes, effects and possible strategies of population growth. – Describe the structure of population in Germany, China and Nigeria	– Show continual desire to contribute to control the effects of rapid population growth	– Population problems of the developing and developed countries – Population growth: Causes, effects and possible strategies Case studies: – Population in Germany – Population in China – Population in Nigeria	– Use geographical documents, internet, maps and photographs analyse the effects of under population, over population and, the population problems of the developing and developed countries. In groups, discuss and compile the findings for a class presentation. – Use the geographical documents, internet, maps and photographs to explain causes and effects of a rapid population growth and describe the control measures of rapid population growth. In groups, discuss and compile the findings for a written report. – In groups, discuss and compile the findings for a class presentation.
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				– In groups, using the geographical documents and internet talking about the types of migration, causes and the effects and the strategies to control Rapid population growth. compile the findings for a class presentation.
Links to other subjects: <i>Economics: Population, Mathematics: Statistics</i>				
Assessment criteria: <i>The learners are able to critically analyse the problems associated with population growth and the strategies of controlling it globally</i>				
Teaching and learning materials: <i>Geographical documents, internet websites, maps and photographs, local environment, atlas, journals, tactile and talking materials, documentary films, population diagrams and statistical graphs.</i>				

Topic Area: Human and Economic Geography		Sub-topic Area: Settlement and urbanisation		
S5 Geography		Unit 10: Rural settlement and Urbanisation in the world		No. of periods: 14
Key unit competence: Evaluate the impact of rural settlement and urbanization on sustainable development in World				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– List the effects of rural settlements on sustainable development of World – Explain the meaning of Basic terms related to urbanization – Identify the functions of urban centres and ports	– Describe the effects of rural settlements on sustainable development of the world – Locate the major world urban centres and ports – Examine the functions of urban centres and ports	– Participate in promoting the awareness of the effects of rural settlement and propose measures for its proper organization – Show concern for the problems caused by the urbanisation on the environment	Rural settlement: – Effects of rural settlements on sustainable development in World Urbanization in the world: – Basic terms related to urbanization – Location of major world urban centres and ports – Functions of urban centres and ports	– Use the geographical documents, textbooks and internet about the effects and solutions of rural settlements. In groups, discuss the findings for a class presentation. – Use the geographical documents, textbooks and internet to explain the concept of urbanisation, terms used, types of urban centres, functions, factors influencing the urban development, the effects of urbanisation on the environment and the measures to control urban problems. In groups, discuss the findings for a class presentation.

– Outline the effects of world urbanization on the environment – Outline the functions of London, Tokyo, and Brasilia cities	– Evaluate the impact of world urbanization on the environment – Describe the functions of London, Tokyo, and Brasilia cities	– Show continual desire to propose strategies to control the problems caused by the urbanisation on the environment	– Impact of world urbanization on the environment – Case studies of cities in developed countries: London and Tokyo – Case studies of the cities in developing countries: Brasilia	– Discuss in pairs the impact of urbanization and suggest solutions possible solutions to the urban problems. Write a report that describes the findings and relevant conclusions. – In groups, describe the functions of London, Tokyo, and Brasilia cities. Write a report that describes the findings and relevant conclusions.
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Links to other subjects: *Entrepreneurship: Urban planning and development*

Assessment criteria: *The learners are able to critically evaluate the impact of rural settlement and urbanization on the sustainable development in the World.*

Teaching and learning materials: *Geographical documents, internet websites, photographs, local environment, atlas, journals, tactile and talking materials, documentary films, urban diagrams and statistical graphs.*

Topic Area: Human and Economic Geography		Sub-topic Area: Economic activities		
S5 Geography		Unit 11: Agriculture in the world		No. of periods: 24
Key unit competence: Analyze the impact of agricultural activities on the sustainable development of different countries.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Outline the types (small holder farming, bush fallowing and shift cultivation), their characteristics, advantages and disadvantages of subsistence crop cultivation. – Outline the types (Cooperative, plantation, irrigation, market gardening and horticulture) their characteristics, advantages and disadvantages of modern crop cultivation	– Describe the types (small holder farming, bush fallowing and shift cultivation) and characteristics of subsistence crop cultivation. – Analyse the advantages and disadvantages of subsistence crop cultivation. – Explain the types (Cooperative, plantation, irrigation, market gardening and horticulture) and characteristics of modern crop cultivation	– Show continual desire to grow different crops using different methods of farming. – Appreciate and contribute to increase the agricultural production.	Types of crop cultivation: – Subsistence/ traditional farming: Types (small holder farming, bush fallowing and shift cultivation), their characteristics, advantages and disadvantages – Modernized farming/ commercial farming: Types (Cooperative, plantation, irrigation, market gardening and horticulture) their characteristics, advantages and disadvantages	– Use geographical documents, textbooks and internet to explain different methods of crop cultivation and their characteristics, their advantages and their disadvantages. In groups discuss and compile the findings for a class presentation. – Use documents, textbooks and internet to describe the factors for increasing agricultural production and the problems affecting agriculture in the developing countries. In groups, discuss and compile the findings and make a class presentation.

– Outline the factors responsible for increasing the agricultural production. – Outline the types, characteristics, advantages and disadvantages of traditional/ subsistence livestock farming – Identify the types, characteristics, advantages and disadvantages of modern livestock farming – Outline the factors and problems affecting livestock farming	– Evaluate the advantages and disadvantages of modern crop cultivation – Explain the factors responsible for increasing the agricultural production. – Explain the types and characteristics of traditional/ subsistence livestock farming – Analyse the advantages and disadvantages of traditional/ subsistence livestock farming – Explain the types and characteristics of modern livestock farming	– Show continual desire to propose strategies/ ways to improve livestock farming.	– Factors that influence agricultural production in the world Case studies we keep only: – Sugar cane grown in South Africa and coffee grown in Brazil, collectivisation/ communes / cooperative farming in China Types of livestock farming: Traditional/ subsistence livestock farming: – Pastoralism: Types (pure nomadism, free Range and transhumance), Characteristics, advantages and disadvantages.	– In groups, use textbooks and the internet to analyse the sugar cane grown in South Africa and coffee grown in Brazil, collectivisation/ communes / cooperative farming in China. Compile your findings in a written report for a class presentation – Use the geographical documents, internet and textbooks to explain the types of livestock farming methods, ways of improving the livestock, factors and problems affecting the livestock. In groups, discuss and compile the findings and make a class presentation.
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– Identify the ways to improve livestock farming	– Evaluate the advantages and disadvantages of modern livestock farming – Explain the factors and problems affecting livestock farming – Identify ways to improve livestock farming – Examine the development of livestock farming in Australia and Holland	Modern livestock farming: – Types (factory farming, ranching and dairy farming), their characteristics, advantages and disadvantages of modern livestock farming – Factors and problems affecting livestock farming and ways of improving livestock farming. Case studies: – Sheep rearing in Australia – Dairy farming in Holland	– In pairs, explain how scientists improve crop yields and combat parasites, and examine how population growth and increased meat demand lead to unsustainable livestock farming practice. Compile your findings in a written report. – In groups, use textbooks and the internet to describe livestock farming activity in Australia and Holland. Compile your findings in a written report for a class presentation.
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Links to other subjects: *Mathematics: Statistics, Economics: Production trends.*

Assessment criteria: *The learners are able to critically analyze the impact of agricultural activities on the sustainable development of different countries*

Teaching and learning materials: *Geographical documents, maps, internet, tactile and talking materials, diagrams, journals, internet and documentaries.*

Topic Area: Human and Economic Geography		Sub-topic Area: Economic activities		
S5 Geography		Unit 12: Forestry in the world		No. of periods: 16
Key unit competence: Evaluate the impact of forestry				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Outline the importance of forestry – Identify the problems that affect forestry – Outline the methods of forest exploitation/ lumbering – Identify the measures for forest conservation	– Evaluate the importance of forestry on sustainable development in the world countries – Analyse the major problems that affect forestry in different countries of World – Describe the methods of forest exploitation/ lumbering – Propose strategies/ measures that can be applied to conserve forests.	– Appreciate the importance of lumbering for sustainable development of the world. – Participate in raising awareness of the dangers of forest over-exploitation and engage in efforts to address them	Lumbering/ forest exploitation: – Importance of forestry – Problems that affect forestry – Methods of forest exploitation/ lumbering – Forest conservation measures. Forestry in selected countries: – Forestry in Gabon – Forestry in British Colombia	– Use internet, textbooks and photographs to find out the different problems and importance of forests of forestry and forest exploitation/lumbering. In groups, discuss the findings and make a class presentation. – Use internet, textbooks and photographs to discuss the methods of forest exploitation, the factors hindering the effective exploitation of the forests and forest conservation measures/strategies. In groups, discuss the findings for a class presentation.

	– Examine the development of forestry in Gabon and British Colombia			– In groups, use textbooks and the internet to describe forestry activities in Gabon and British Columbia. Compile your findings in a written report for a class presentation.
Links to other subjects: <i>Biology: Botany (plants) and ecology.</i>				
Assessment criteria: <i>The learners are able to critically evaluate the impact of forestry on sustainable development in the world countries.</i>				
Teaching and learning materials: <i>Geographical documents, photographs, maps, local environment, tactile and talking materials, journals, internet websites/jaws software and newspapers.</i>				

Topic Area: Human and Economic Geography			Sub-topic Area: Economic activities	
S5 Geography		Unit 13: Fishing in the world		No. of periods: 17
Key unit competence: Analyze the impact of fishing on the sustainable development of different countries				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the major fishing grounds of the world. – Outline different methods used in fishing – Name the major types of fish/ fish species in the world. – Outline the factors influencing development of fishing in the world	– Locate on the map of world the major fishing grounds – Explain the methods used in fishing – Explain the types of fish/fish species in the world. – Analyze the factors influencing development of fishing in the world – Evaluate the importance of fish and fishing.	– Show commitment to advise fisheries about sustainable method that can be used to develop inland and marine fishing – Appreciate the importance of fishing to human life and economic development of the world.	– The major fishing grounds of the world. – Methods used in fishing – Types of fish/ fish species in the world. – Factors influencing development of fishing in the world – Importance of fish and fishing.	– Use internet, the geographical documents, fishing maps and photographs to explain the major fishing grounds, methods used for fishing and the types of fishes caught. In groups, discuss and compile the findings and make a class presentation. – Use the geographical documents and internet to analyse the factors favouring the development of fishing, importance of fish and fishing, problems and solutions associated with fishing. Do group discussions on the findings and make a class presentation.

– Identify the importance of fish and fishing. – Outline the problems affecting marine and inland fisheries – Outline the possible solutions to the problems affecting fishing – List the methods used for fish conservation and fish preservation	– Explain the problems affecting marine and inland fisheries and propose possible solutions – Propose fish conservation and preservation methods that support sustainable fishing – Explain the fishing development in the selected countries Japan: (Northwest Pacific) and Morocco (West Africa).	– Show continual desire to participate in solving the problems affecting fishing and advocate for fish conservation measures.	Marine fishing: – Problems affecting marine fisheries and possible solutions Inland fishing: – Problems and prospects inland fisheries – Fish conservation and preservation methods Case studies: – Japan (Northwest Pacific) – Morocco (West Africa).	– In groups, use textbooks and internet to explain fish conservation and preservation methods and compile the findings and make a group presentation. – In groups, use internet, textbooks and other geographical documents to analyse critically the fishing in Norway, Japan, Peru or Morocco for discussions in reference to the knowledge acquired in the current unit and write a report on the findings. – In pairs, discuss the impact of overfishing, and how to address those impacts. Write a report that summarizes the relevant conclusions.
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Links to other subjects: *Biology: Aquatic life (zoology)*

Assessment criteria: *The learners are able to critically analyze the impact of fishing on the sustainable development of different countries.*

Teaching and learning materials: *Geographical documents, photographs, local environment, internet websites/jaws software, statistical diagrams and graphs, journals, tactile and talking materials.*

Topic Area: Human and Economic Geography		Sub-topic Area: Economic activities		
S5 Geography		Unit 14: Mining in the world		No. of periods: 16
Key unit competence: Examine the impact of mining on sustainable development of different countries of the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the distribution of minerals in the world. – List the methods used for mining in the world. – State the factors affecting the mineral exploitation. – Outline the effects of mining to the economies of the world.	– Locate the major minerals of the world on a map. – Explain the methods used for mining. – Describe the factors affecting the mineral exploitation. – Examine the effects of mining to the economy of a country.	– Show continual desire to protect the natural environment as it plays a big role in the distribution of minerals across different regions of the world. – Appreciate the significance of minerals for the sustainable development of various regions worldwide.	– World distribution of major minerals – Methods used in mining – Factors favouring mineral exploitation in the world – The effects of mining to the economies of the world.	– Use internet and textbooks, to explain and locate the major minerals of the world, methods used for mining, factors affecting mineral exploitation and the effects of mining. In groups, discuss the findings for a class presentation. – In groups, use internet, textbooks and other geographical documents to analyse mining in Russia, China, Nigeria, and South Africa. Compile findings and write a report on the findings.

– Explain mining in relation to the selected countries (South Africa, China, Nigeria and Russia).	– Apply the knowledge to evaluate the role played by minerals in the sustainable development of different economies of the world.		Case studies: – Mining in South Africa – Mining in China – Mining in Nigeria – Mining in Russia	
Links to other subjects: <i>Chemistry: Transitional metals and chemical properties of minerals.</i>				
Assessment criteria: <i>The learners are able to critically Examine the impact of mining on sustainable development of different countries of the world.</i>				
Teaching and learning materials: <i>Geographical documents, maps, tactile and talking materials, atlases, journals, photographs, local environment, internet/Jaws software and samples of minerals.</i>				

SENIOR SIX

6.4. GEOGRAPHY FOR SENIOR 6

6.4.1. General competences of geography at the end of senior six:

- Explain different physical geographical phenomena and their impact on sustainable development in the world.
- Explain different problems associated with the physical, human and economic environment of the world and suggest solutions.
- Evaluate the major challenges related to the exploitation of resources in the world.
- Compare different modes of development in the world.
- Interpret physical and human features on the maps and the photographs.

6.4.2. Syllabus units for senior six

Unit	Unit title	Periods
1	Cross section and sketch maps	8
2	Interpretation of photographs	7
3	The origin and distribution of the continents	12
4	External landform processes and related features	32
5	Wave erosion and deposition	21
6	Rocks and minerals	14
7	Classification of soil and soil formation	7
8	Climate change	9
9	Global drainage systems	32
10	Power and Energy production in the world	14
11	Industrialization in the world	13
12	Transport and communication in the world	14
13	Trade and commerce in the world	14
14	World multipurpose river projects	10
15	Environmental conservation and tourism	24

GEOGRAPHY SYLLABUS FOR SENIOR SIX

Topic Area: Practical Geography		Sub-Topic Area: Map work interpretation		
S6 Geography		Unit 1: Cross sections and Sketch maps		No. of periods: 8
Key unit competence: Draw cross sections and sketch maps by reduction or enlargement.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the contours on a given map – Explain the meaning of the vertical exaggeration, the gradient, amplitude and inter visibility – Identify the factor number for enlargement and reduction of the given map	– Analyse the appearance of the contours on the topographic map. – Draw cross sections using contours. – Calculate the vertical exaggeration, vertical interval, gradient and the amplitude of relief.	– Demonstrate an attitude of using the topographic map and the cross sections to interpret relief of a given area. – Demonstrate positive attitude of drawing cross sections of the hilly terrain represented on a topographic map.	– Cross section or sketch section – Determining the vertical exaggeration, the gradient, amplitude and inter visibility – Drawing a sketch map – Enlargement and reduction of the map	– In groups, use topographic maps to identify the contours lines, determine the vertical interval, calculate the vertical exaggeration, vertical interval, the gradient, the amplitude and show inter-visibility of given topographic map. Compile the findings and share them to the class. – Construct a cross-section by identifying and drawing the section line, marking and transferring contour points, plotting heights using the vertical scale, and smoothly joining and labeling the features using the given scale. Present the generated relief feature to the class.

	– Determine the inter-visibility on the maps. – Determine the scale of the sketch map. – Make a sketch map of the reduced or enlarged map.	– Show consistent commitment to protecting hilly areas shown on a map.		– In pairs, enlarge and reduce a map by identifying the given scale, calculating the new scale, redrawing the map, adjust the key features according to the new measurements, and present the final enlarged or reduced map to the class for comparison and discussion.
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Links to other subjects: *Mathematics: Drawing of the graphs (x and y axes)*

Assessment criteria: *The learners are able to correctly draw the cross sections and sketch maps by reduction or enlargement.*

Teaching and learning materials: *Sample topographic maps, tactile or talking maps drawing equipment, magnifying lenses, local environment, Brail drawing materials/equipment, mathematical instruments.*

Topic Area: Practical Geography		Sub-Topic Area: Photographic interpretation		
S6 Geography		Unit 2: Interpretation of Photographs		No. of Periods:7
Key unit competence: Interpret physical and human features on photographs				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of a photograph. – Identify the different types of photographs. – Identify the physical and human aspects on a photograph. – State the sections of a photograph. – Identify different techniques for drawing a sketch of a photograph by reduction and enlargement.	– Analyse different types of photographs. – Interpret physical and human aspects represented on a photograph. – Describe the different sections of a photograph. – Draw a sketch photograph using the given scale and factor number to reduce or enlarge it.	– Appreciate the importance of the photographs to interpret human and physical features represented on a photograph. – Appreciate the relationship between the physical and human aspects represented on photographs.	– Meaning and types of photographs – Sections of photographs – Interpretation of physical and human aspects in photographs – Drawing sketches of photographs by reduction and enlargement	– In groups, use photographs, geographical documents, diagrams and internet to analyse the types of photographs, sections of a photograph and create a sketch photograph by reduction or enlargement. Make a class presentation to share the findings.

– State the relationship between the physical and human aspects on the photographs.	– Analyse the relationship between the physical and human aspects on the photographs.		– The relationship between physical and human aspects on photographs	– Use the given photographs to identify the physical and human aspects on the photographs and their inter-relationships. In groups, discuss the findings, write an essay report to present in a class.
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Links to other subjects: *Mathematics: Geometry (enlargement), point lines and angles*

Assessment criteria: *The learners are able to correctly interpret the physical and human aspects on the photographs and draw sketch photos by reduction or enlargement of the photographs.*

Teaching and learning materials: *Geographical documents, photographs, cameras, internet, drawing equipment, magnifying lenses, tactile and talking materials.*

Topic Area: Physical Geography		Sub-Topic Area: Understanding the earth and universe		
S6 Geography	Unit 3: The origin and distribution of the continents			No. of periods: 12
Key unit Competence: Examine the origin and the distribution of the continents				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the theories of origin and distribution of the continents and ocean basins. – Outline the evidences of the continental drift. – State the major types of plate boundaries and tectonic plates. – Identify the effects of tectonics plate on the landscape.	– Describe the origin and distribution of the continents and ocean basins. – Examine the formation, shapes and the current positions of the continents and ocean basins. – Justify the view that the continents drifted.	– Appreciate the origin and the distribution of the continents and ocean basins. – Develop curiosity and responsibility in understanding the causes and effects of continental drift and the distribution of continents and ocean basins.	– Theories of the origin and distribution of continents and ocean basins. – Evidence of continental drift – Effects of continental drift on the evolution of physical features – Concept of Plate Tectonics – Major plates and effects of plate tectonics – The theory of Isostasy	– Use the geographical documents, maps, photographs and internet to analyse the theories of the origin of the continents and ocean basins, their distribution, evidence of continental drift and the effects of continental drift on the landscape. In groups, discuss and write down the findings for a class presentation.

– State the main theories of Isostasy	– Explain the effects of the continental drift on the landscape. – Examine the causes of continental drift and the global distribution of continents. – Describe the main theories of Isostasy			– Use the geographical documents, maps and internet to describe the theories of isostasy, tectonic plates; the characteristics of tectonic plates and its effects on the land. In groups, discuss and compile the findings and make a class presentation.
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Links to other subjects: *Physics: The forces of compression and tension.*

Assessment criteria: *The learners are able to correctly examine the theories of the origin and the distribution of the continents.*

Teaching and learning materials: *Geographical documents, maps, globe, journals, tactile and talking materials, atlases, photographs and internet/Jaws software.*

Topic Area: Physical Geography`		Sub-Topic Area: Landform evolution and processes		
S6 Geography		Unit 4: External landform processes and related features		No. of periods: 36
Key unit competence: Analyze the different features resulting from external processes and their relationships with human activities.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify different types and processes of weathering. – State the major factors influencing the weathering. – Outline the interdependence of chemical and physical weathering processes in different regions	– Describe different weathering processes. – Examine the factors affecting the weathering. – Evaluate the interdependence between the physical and the chemical weathering processes in different regions.	– Appreciate how landforms formed by weathering contribute to economic development. – Show a continual desire to analyze the causes and effects of mass wasting and to propose appropriate mitigation measures.	– Meaning, types and processes of weathering – Differences and similarities between physical and chemical weathering – Factors influencing weather – Interdependence of chemical and physical weathering in different regions. – Weathering in limestone regions	– Use documents, internet, maps, photographs or the local environment to identify different types and processes of weathering and the factors influencing weathering. In groups, discuss the findings and make a class presentation. – Use textbooks, internet and photographs to describe the different features resulting from the weathering in different regions (limestone regions, humid tropical regions, desert and semi desert areas).

– Identify the major features resulting from the weathering in the Karst regions. – Outline the features resulting from the weathering in the humid tropical and the desert areas. – Explain the processes of weathering in the glaciated areas and their related features. – Identify the different types of mass wasting in different areas, their causes and the effects.	– Explain the major features resulting from the weathering in the Karst regions. – Describe the major features resulting from weathering in the humid tropical and the desert regions. – Analyse the processes of weathering in the glaciated areas and the associated landforms. – Evaluate the causes and the effects of mass wasting.	– Appreciate the relationship between different weathering landforms and the human activities.	– Weathering in humid tropical and desert region – Weathering in the glaciated cold regions – Factors for the formation and movement of glaciers – The work of glaciers/glaciation (erosion, transportation and deposition) and their resultant landforms – Impact of glaciation on human activities – Mass wasting: Meaning and types – Causes, effects and measures to control mass wasting	– In groups, discuss and compile the findings for a class presentation. – Use textbooks, internet and photographs to analyse the weathering in the glaciated areas (causes of glacial formation action of glaciers, and the landforms associated with glaciation). In groups, discuss and write down the findings for a class presentation. – Use geographical documents, the internet, maps, and photographs to examine the types of mass wasting, their causes, effects, and control measures. In groups, discuss and compile the findings for a class presentation.
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– Identify the relationship between weathering landforms and the human activities.	– Examine the relationship between weathering landforms and the human activities		– The relationship between weathering landforms and human activities	– Use the geographical documents and internet to analyse the relationship between weathering and human activities. In groups, discuss and compile the findings for a class presentation.
Links to other subjects: <i>Physics: force/pressure; Chemistry: solubility/solvent action.</i>				
Assessment criteria: <i>The learners are able to correctly analyze the different features resulting from external processes and their relationships with human activities.</i>				
Teaching and learning materials: <i>Geographical documents, maps, photographs, journals, tactile and talking materials, local environment, charts, internet website and fieldwork materials.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Landform evolution and processes		
S6 Geography		Unit 5: Wave erosion and deposition		No. of periods: 21
Key unit competence: Evaluate the different landforms resulting from wave action and their relationships with human activities.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of concepts of Wave, coast, shore and longshore drift – Explain the types of waves (constructive and destructive) – Identify factors determining the strength of waves – Identify the wave processes of waves (erosion, transportation and deposition) and their associated landforms	– Describe the types of waves – Explain the factors determining the strength of waves – Analyse the wave processes (erosion, transportation and deposition) and their associated landforms – Explain the factors influencing the formation of coastal landforms	– Show a continual desire to protect the environment along different types of coasts. – Appreciate the importance of the different landforms created by the wave erosion and the deposition.	– Meaning of key concepts (Wave, coast, shore, longshore drift – Types of waves (constructive and destructive) – Factors determining the strength of waves – Wave action/ processes of waves (erosion, transportation and deposition) and their associated landforms – Factors influencing the formation of coastal landforms	– Use geographical documents, internet, maps, diagrams and photographs to examine the processes of wave erosion, wave deposition, factors influencing the formation of coastal landforms, landforms resulting from wave erosion and deposition and the importance of coastal landforms.

– Outline the factors influencing the formation of coastal landforms – Outline importance of landforms produced by wave action – Identify the types of coasts – Explain the nature and types of coral reefs – Explain the process of coral reefs formation – Identify the conditions influencing coral formation – Identify the effects and problems related to coral reefs	– Examine the importance of landforms produced by wave action – Describe the types of coasts – Explain the conditions influencing coral formation – Examine the effects and problems related to coral reefs – Distinguish between isostatic from Eustatic changes on the coast – Describe the types of sea level changes – Explain the causes and features resulting from sea level changes	– Show the continual desire to propose the mitigation measures to the environment affected by isostatic and eustatic changes on the coast	– Importance of landforms produced by wave action Types of coasts: – Submerged upland coast, submerged lowland coast and Emerged coasts Coral reefs: – Nature and types Formation of coral reefs: – The process of coral reefs formation – General conditions influencing coral formation Case study: – Great Barrier Reef in Australia – Impact and problems related to coral reefs – Isostatic and Eustatic changes on the coast	– In groups, discuss and compile the findings for the class presentation. – Use documents, internet, maps, diagrams and photographs analyse types of coasts and coral reefs, formation of coral reefs, factors influencing coral reefs and impact of coral reefs. – In groups, discuss and compile the findings for the class presentation. – Use documents, internet, maps, diagrams and photographs to research on the sea level changes, types, causes, associated features, evidences, effects.
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– Explain the meaning and types of sea level changes – Outline the causes and features resulting from sea level changes – Identify the effects of sea level changes on human activities – Identify the marine relief features.	– Examine the effects of sea level changes on human activities – Describe the marine relief features		Sea level change: – Meaning and types of sea level changes – Causes of sea level changes – Features resulting from sea level changes – Effects of sea level changes on human activities Marine Relief: – Continental shelf – Continental slope – Deep sea plains – Oceanic trenches – Ocean ridge	– In groups, discuss and compile the findings for a class presentation and draw conclusions
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Links to other subjects: *Physics: oscillations, gravitational force and power.*

Assessment criteria: *The learners are able to critically evaluate the different landforms resulting from wave action and their relationships with human activities.*

Teaching and learning materials: *Geographical documents, maps, atlas, journals, photographs, diagrams, talking and tactile materials, local environment, charts and internet.*

Topic Area: Physical Geography		Sub-Topic Area: Rocks and Minerals		
S6 Geography		Unit 6: Rocks and Minerals		No. of periods: 14
Key unit competence: Evaluate economic importance of rocks and minerals				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the various types of rocks in the world and their characteristics. – Identify different components of the rocks. – Identify the impact of rocks on the landscape and human activities – Identify the various types of minerals in the world.	– Examine different types of rocks and their characteristics. – Explain the various components of the rocks. – Evaluate the impact of rocks on the landscape and human activities – Describe the physical and the chemical properties of the rocks.	– Appreciate the economic importance and uses of rocks and minerals. – Show commitment to the sustainable use of rocks and minerals to protect the natural environment."	– Meaning, types and characteristics of rocks – Composition of rocks – Physical and chemical properties of rocks – Impact of rocks on the landscape and human activities – Meaning and types of minerals/ores	– Use textbooks, internet and photographs to identify the types, characteristics, composition and importance of rocks. Write down the findings. In groups, discuss and compile the findings for a class presentation. – Use textbooks, the internet, and photographs to identify different types of minerals and describe their physical and chemical properties. – In groups, discuss and compile the findings for a class presentation.

– Identify the physical and the chemical properties of the minerals. – Identify the industrial products extracted (manufactured) from the minerals and their economic use.	– Describe different types of minerals. – Explain the physical and the chemical properties of the minerals. – Evaluate the importance of the industrial products extracted from the minerals.		– Physical and chemical properties of minerals – Uses of minerals and manufactured products	– Carry out a field trip in any one of the mining sites; find out the characteristics of the different minerals present in this site and write down the findings for group discussion and a class presentation. – Using the internet and textbooks to research the industrial value of the minerals and the manufactured products. Write down the findings for a class presentation.
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Links to other subjects: *Chemistry: chemical properties.*

Assessment criteria: *The learners are able to correctly to evaluate the economic importance of rocks and minerals.*

Materials: *Geographical documents, maps, atlases, tactile or talking materials, photographs, local environment, internet /Jaws software, mining sites for field study, rock and mineral samples.*

Topic Area: Physical Geography		Sub-Topic Area: Soils		
S6 Geography		Unit 7: Classification of soils and soil formation		No. of periods: 7
Key unit competence: Examine the factors and the processes responsible for the formation of different types of soils.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the examples of zonal, azonal and inter-zonal soils. – Identify the major types of soils in the world. – Identify the processes of the soil formation. – List the factors responsible for the soil formation.	– Classify the major types of soil in the world. – Evaluate the distribution of the major types of soil in the world. – Describe the processes of the soil formation. – Explain the factors favouring the formation of the soils.	– Appreciate the economic importance of the soils in the world. – Show respect for the soil conservation measures in the world.	– Classification and distribution of the major types of soil in the world (Zonal, Azonal, Intrazonal soils). – Processes of soil formation – Factors of soil formation – Soil erosion: Areas of severe soil erosion	– Use internet, geography textbooks, maps of soils and photographs to research on the classification, the types of soils and their distribution, the processes and the factors favouring the formation of the soils. In groups, discuss and compile the findings for a class presentation. – Use internet, a range of textbooks, maps and photographs to research on the causes, the effects of soil erosion, and the measures for soil conservation. In groups, discuss and compile the findings for a class presentation and draw the relevant conclusions.

– Identify the areas of severe soil erosion	– Explain the causes and the effects of the soil erosion in the world. – Locate the areas of severe soil erosion			– In groups, discuss the economic importance of the soils and make a class presentation. Also engage in active Q&A session.
Links to other subjects: <i>Biology: decomposition and humus formation</i>				
Assessment criteria: <i>The learners are able to critically examine the factors and the processes responsible for the formation of different types of soil.</i>				
Teaching and learning materials: <i>Geographical documents, maps, atlases, journals, tactile or talking materials, photographs, local environment, internet /Jaws software and local environment.</i>				

Topic Area: Physical Geography		Sub-Topic Area: Weather and climate		
S6 Geography		Unit 8: Climate change		No. of periods: 9
Key unit competence: Examine the impacts of climate change in Rwanda and other countries.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of climate change. – State the causes of climate change. – Identify the effects of climate change. – Locate the areas most affected by the climate change in the world. – Identify the causes and the effects of global warming and green house.	– Explain the causes of climate change. – Evaluate the effects of climate change. – Explain the causes and the effects of global warming and green house. – Examine the ways of adaptation and mitigation to climate change.	– Appreciate the importance of adaptation and mitigation on climate change. – Show concern for the man's contribution towards the climate degradation. – Show concern for the impact of climate change.	– Meaning and causes of climate change – Effects of climate change worldwide (Global, Africa, Rwanda) – Meaning and causes of global warming and green house phenomena – Effects of global warming and greenhouse phenomena	– Use the geographical documents, internet, maps, and photographs to identify the causes and the effects of climate change; measures of the climate adaptation and write down the findings. In groups, discuss and compile the findings for a class presentation. – In groups, discuss the measures for mitigating the climate change and make a class presentation

– Outline the measures of the climate adaptation and mitigation. – Identify the causes and the effects of desertification.	– Examine the causes and the effects of desertification.	– Show concern for the impact of global warming and green house effects.	Adaptation and mitigation measures for climate change: – Adaptation measures – Mitigating measures Desertification: – Meaning and causes of desertification – Effects of desertification	– Use the geographical documents, internet, maps and photographs to research about the causes and the impact of global warming and the greenhouse effect. In groups, use the geographical documents, internet, maps and photographs to research on the causes and the effects of the desertification. – Write down the findings for a class presentation.
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Links to other subjects: *Chemistry: The greenhouse gases*

Assessment criteria: *The learners are able to critically examine the causes and the effects of climate change and propose ways of adaptation and mitigation to climate change.*

Teaching and learning materials: *Documents, brailled textbooks, maps, atlases, tactile or talking materials, photographs, internet /Jaws software and local environment.*

Topic Area: Physical Geography		Sub-Topic Area: Drainage		
S6 Geography		Unit 9: Global Drainage Systems		No. of periods: 32
Key unit competence: Examine the economic importance of global drainage systems and the reasons for their conservation.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of river discharge, velocity, divide, basin, width, gradient/slope and catchment area) – State the different types of rivers. – Identify the river processes involved in river erosion, transportation and deposition – Identify the characteristics of river profiles (cross and long river profiles)	– Describe the different types of rivers. – Analyse the river processes involved in river erosion, transportation and deposition – Explain the characteristics of river profiles (cross and long river profiles)	– Appreciate the importance of the global drainage. – Show concern for the causes and the effects of the ocean currents and the tides and propose their mitigation measures.	– Meaning of a river and the associated terms – The types of rivers (intermittent, perennial and ephemeral) – The work of a river (erosion, transportation and deposition) – River profiles (cross and long river profile) and their characteristics The landforms associated with a long river profile: – Formation of landforms in youthful stage (waterfalls, rapids, potholes, plunge pool and interlocking spurs)	– Use geographical documents, maps, photographs and internet to explore the river profile and its associated landforms. In groups, discuss the findings for a class presentation.

– Mention the landforms associated with youthful stage, mature stage and lower/old stage. – Explain the meaning of river capture – Identify the causes of river capture – Identify the features associated with river capture and their effects. – Explain the meaning of river rejuvenation – Identify the causes of river rejuvenation – Identify the effects of river rejuvenation on landscape – Outline the different types of river drainage patterns.	– Describe the landforms associated with youthful stage, mature stage and lower/old stage. – Examine the causes of river capture – Describe the features associated with river capture and their effects. – Explain the causes of river rejuvenation – Describe the effects of river rejuvenation on landscape – Illustrate the different types of river drainage patterns.	– Show continual desire to protect the water bodies for the conservation of the environment. – Participate actively in promoting the responsible and sustainable use of water bodies and their connected landforms.	– Formation of landforms in mature stage (meanders, gorges and canyons, alluvial fans, river benches and river terraces) – Formation of landforms in lower/old stage (developed meanders, ox-bow lakes, flood plains, braided channels, deltas, estuaries and levees) River capture – Meaning of river capture – Causes of river capture – Features associated with river capture – Effects of river capture River rejuvenation – Meaning of river rejuvenation – Causes of river rejuvenation	– Use textbooks, diagrams, photographs and internet, research about the landforms resulting from the river capture and river rejuvenation. In groups, discuss and compile the findings for a class presentation. – In groups, use geographical textbooks or research on internet and describe the different drainage systems. Compile the findings for a class presentation.
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– Explain the meaning of superimposed and antecedent drainage – Identify the impact of rivers on landscape and human activities – Name the main major rivers of the world – Identify the types of lakes and their mode formation – Name the seas and oceans of the world – Identify the importance of lakes, Seas and Oceans – Explain the meaning and characteristics of ocean currents – Identify types of ocean currents	– Describe the superimposed and antecedent drainage – Examine the impact of rivers on landscape and human activities – Locate on a map of the world the main major rivers of the world – Categorize the types of lakes according to their mode formation – Locate on the map of world the seas and oceans of the world – Explain the importance of lakes, Seas and Oceans – Describe the characteristics of ocean currents		– Effects of river rejuvenation on landscape Drainage patterns – Trellis, dendritic, rectangular, radial, centripetal, annular, barbed drainage pattern, pinnate, herringbone drainage pattern and parallel drainage pattern – Superimposed and antecedent drainage – Impact of rivers on landscape and human activities – Major rivers of the world Lakes, Seas and oceans: – Types of lakes and their mode formation – Distribution of seas and oceans – Importance of lakes, Seas and Oceans	– Use the geographical documents and a drainage map of the world to identify the major lakes according to their mode of formation, the impact of lakes; seas and oceans, the types, causes and the effects of the tides, and their characteristics, the effects of ocean currents on the adjacent lands. In groups, discuss and compile the findings for a class presentation.
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– Identify the causes / factors of ocean currents – Identify the effects of ocean currents on climate and adjacent lands – Explain the meaning of tides – Identify the types, causes and effects of tides – Identify the reasons of protecting global water bodies in the World	– Examine the effects of ocean currents on climate and adjacent lands – Explain the types, causes and effects of tides – Explain the reasons of protecting global water bodies in the World – Locate on the map of world different types of ocean currents – Explain the causes/factors of ocean currents		Ocean currents: – Meaning and characteristics of ocean currents – Types and location of ocean currents – Causes / factors of ocean currents – Influence of ocean currents on climate and adjacent lands Tides: – Meaning and types – Causes and effects of tides – Reasons for protection of global water bodies	– In groups, use internet and geographical documents to examine the importance of the global water bodies. Compile the findings for a class presentation.
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Links to other subjects: *Chemistry: Solution and salinity; Physics: velocity, volume, depth and oscillation.*

Assessment criteria: *The learners are able to critically examine the economic importance of global drainage systems and the reasons for their conservation*

Teaching and learning materials: *Documents, brailled textbooks, maps, atlases, tactile or talking materials, photographs, internet /Jaws software and local environment.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S6 Geography	Unit 10: Power and Energy Production in the world			No. of periods: 14
Key unit competence: Evaluate the impact of power and energy production on sustainable development of the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify the major world's sources of energy. – Explain the factors affecting the power production. – Explain the importance of the power and energy in the world.	– Distinguish with examples the major energy sources used in the world. – Describe the factors affecting the power production in the world. – Evaluate the importance of the power and energy on sustainable development of the world.	– Show concern for the low levels of power and energy production in the world and propose the strategies/ measures to increase it. – Demonstrate continual desire to use renewable sources of the power and energy that don't affect negatively the environment.	Classification of energy sources in the world: Non- renewable energy sources: – Nuclear energy (Uranium) – Coal (Types and uses of coal) – Petroleum Oil (Three grades of crude oil according to gasoline yields and uses of petroleum) – Natural gas Renewable energy sources – Wind energy	– Use the geographical documents, maps, photographs, internet, video clips or the local environment to identify the different sources of power and energy used in the world, the factors favouring the power production, the importance of power in the development and the problems hindering the development of the power and energy. In groups, discuss and compile the findings for a class presentation.

– State the problems hindering the development of the power and energy and suggest the solutions.	– Explain the problems hindering the development of the power and energy and propose the possible solutions.	– Appreciate the importance of the power and energy on the sustainable development of the world's economy. – Show respect for the efforts in the development of power and energy in the world.	– Water energy (Physical and economic factors favouring hydroelectric power production) – Solar energy – Biomass (Wood fuel, Waste products (Biogas) and gasohol) – Geothermal energy – Tidal energy – Factors and importance of power and energy production in the world. – Factors favoring power and energy production in the World. – Importance of power and energy in the sustainable development of the world. – Problems and possible solutions for power and energy – Problems hindering the development of power and energy production in the world.	– In groups, discuss the possible solutions to the problems hindering the development of power and energy and make a class presentation. – Our reliance on the decreasing stock of fossil fuels continues to rise, while global energy demand is also increasing and the production of renewable energy is growing very slowly. As a world, how can we manage this impending energy crisis? In groups, study the above scenario and compile relevant conclusions and recommendations to share with your classmates during a class presentation
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			– Possible solutions for power and energy production in the World	
Links to other subjects: <i>Physics: Power and energy.</i>				
Assessment criteria: <i>The learners are able to critically evaluate the impact of power and energy production on sustainable development of the world.</i>				
Teaching and learning materials: <i>Geographical documents, brailled textbooks, maps, atlases, tactile or talking materials, photographs, internet /Jaws software and local environment.</i>				

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S6 Geography		Unit 11: Industrialisation in the world		No. of periods: 13
Key unit competence: Examine the impact of industrialization on sustainable development of the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Identify different types of industries. – State the factors influencing the location of the industries. – Name the major industrial regions of the world. – Give the importance of the industrial development in the world.	– Classify different types of industries according to their products. – Evaluate the factors influencing the location of the industries. – Locate the major industrial regions of the world. – Evaluate the importance of the industrial development in the world.	– Appreciate the contribution of industries on sustainable development. – Show concern for the effects of industrialisation and demonstrate the continual desire to protect the environment	– Meaning and classification of industries – Factors influencing location of industries – Major world industrial regions – Importance of industries and problems affecting industrial development – Problems resulting from industrial development and ways to overcome them.	– In groups, discuss about the classification of the industries and their types of industrial products. – Use the geographical documents, internet and photographs and discuss the factors influencing the location of industries and industrial development, – In pairs, use maps and geography textbooks to locate the major world industrial regions. Share the findings to classmates during the class presentation.

– Identify the problems affecting the industries and the effects of industrial development. – Identify the problems resulting from industrial development and ways to overcome them	– Evaluate the problems affecting the industries and the effects of the industrial development in the world and suggest the solutions. – Describe the industrial development of Egypt, USA and China		– Case studies on major world industrial regions: Egypt, USA and China	– In groups, search from internet and textbooks the importance of industrial development, problems affecting and resulting from industrial development. In groups, discuss and compile the findings to make a class presentation. – Use documents, maps, internet and photographs to describe the industrialisation in the selected countries (Egypt, USA and China). In groups, discuss the findings and write a report.
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Links to other subjects: *History: Industrial revolution*

Assessment criteria: *The learners are able to critically examine the impact of industrialization on sustainable development of the world*

Teaching and learning materials: *Geographical documents, brailled textbooks, maps, atlases, tactile or talking materials, photographs, internet /Jaws software and local environment.*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S6 Geography	Unit 12: Transport and Communication in the world		No. of periods: 14	
Key unit competence: Analyze the impact of transport and communication on sustainable development in the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of transport and communication. – Identify the major types of transport. – Outline the advantages and disadvantages of the different types of transport – Identify the factors influencing the development of the different types of transport.	– Describe the major types of transport. – Explain the factors influencing the development of the different types of transport. – Explain the advantages and disadvantages of the different types of transport. – Evaluate the importance of transport.	– Appreciate the importance of the different types of transport. – Use the available resources like the inland waterways, airports, roads. to compare the transport in Rwanda with the other countries.	– Transport: – Meaning and types of transport – Advantages and disadvantages of each type of transport. – Factors influencing the development of transport – Importance of transport	– Use the geographical documents, internet, maps, photographs explore the different types of transport, the factors influencing the development of the transport, the advantages and the disadvantages of each type of transport, the importance of transport and the challenges facing the different types of transport. In groups, discuss and compile the findings for a class presentation.

– Give the importance of transport. – Outline economic importance, and regional significance of the Lawrence Seaway, Tanzam Railway and Trans-African Highway – State the different types of communication. – Give the importance of communication. – Identify the problems affecting the communication.	– Analyse the problems/ challenges affecting transport and propose the strategies to improve transport in the world – Examine the economic importance, and regional significance of the Lawrence Seaway, Tanzam Railway, and Trans-African Highway – Explain the different means of communication. – Evaluate the importance of communication.	– Appreciate the advantages and the disadvantages of all types of transport and communication. – Show concern for the challenges encountered in the transport and communication and the desire to improve the transport and communication means.	– Problems/ challenges affecting transport and strategies for improving transport. – Case studies on transport: St. Lawrence Sea way and Tanzam railway and Trans- African Highway. Communication: – Meaning and means of communication – Importance of communication – Problems affecting communication	– Use the geographical documents, internet, maps and photographs, explain the air, water, road and railway transport in the selected countries of Africa and North America as examples. In groups, discuss, compile the findings and write a report for class presentation. – Use the geographical documents, internet, maps and photographs or the local environment to research on the different means of communication, the importance of communication and the problems affecting communication. In groups, discuss and compile the findings for a class presentation.
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	– Explain the problems affecting the communication and propose the solutions.		– Possible solutions to the problems affecting communication	– In pairs, use internet and textbooks and propose the possible solutions to the problems affecting communication in the world.
Links to other subjects: <i>Economics: Economic sectors (Transport and Communication).</i>				
Assessment criteria: <i>The learners are able to critically analyze the impact of transport and communication on sustainable development in the world</i>				
Teaching and learning materials: <i>Geographical documents, brailled textbooks, maps, atlases, tactile or talking materials, photographs, internet /Jaws software and local environment.</i>				

Topic Area: Human and Economic Geography			Sub-Topic Area: Economic activities	
S6 Geography		Unit 13: Trade and Commerce in the world		No. of Periods:14
Key unit competence: Evaluate the impact of trade and commerce on sustainable development in the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Explain the meaning of trade and commerce. – Identify the different types of trade and commerce. – Outline the factors influencing the international trade. – Give the importance of the international trade. – Identify problems affecting international trade in developing countries	– Explain different types of trade – Describe the factors influencing the trade. – Evaluate the importance of international trade in development – Examine the problems affecting international trade in developing countries	– Appreciate the importance of trade to the sustainable development of the World. – Show the continual desire to improve trade.	– Meaning and types of trade – Factors influencing international trade – Importance of international trade in development – Problems affecting international trade in developing countries – Possible solutions to the problems affecting international trade in developing countries	– Use internet, maps and a geography textbook, on different types of trade, factors influencing trade, importance of international trade, major financial centres of the world, world trading blocks and the regional integration. In groups, discuss and compile the findings for a class presentation.

– Outline possible solutions to the problems affecting international trade in developing countries – State the major financial centres of the world. – Identify the major trading patterns and the world trading blocks. – Outline the factors and problems affecting regional integration and possible solutions – Outline the advantages of regional integration and trading blocs such as EAC, ECOWAS, OPEC and EU.	– Propose possible solutions to the problems affecting international trade in developing countries. – Locate major financial centres of the world. – Describe the major trading patterns and the trading blocks of the world. – Examine the advantages of regional integration and trading blocs such as EAC, ECOWAS, OPEC and EU.	– Appreciate the importance of the major financial centres and trading blocks. – Appreciate the importance of the regional integration.	– Major financial centres. – Trading blocks and regional integration: meaning, advantages and disadvantages. – Factors and problems affecting regional integration and possible solutions. – Case studies on regional integration and trading blocs (EAC, ECOWAS, OPEC & EU).	– In groups, use textbooks and internet and conduct a discussion on the problems affecting the trade and propose possible solutions to mitigate those problems. Compile the findings for a class presentation. – Use documents, maps, internet and photographs to examine the advantages of regional integration and trading blocs such as EAC, ECOWAS, OPEC and EU. In groups, discuss the findings and write a report.
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Links to other subjects: *Mathematic: Statistics and the frequency.*

Assessment criteria: *The learners are able to critically evaluate the impact of trade on the sustainable development of the world.*

Teaching and learning materials: *Geographical documents, internet, photographs, maps, local environments, tactile and talking materials and journals*

Topic Area: Human and Economic Geography		Sub-Topic Area: Economic activities		
S6 Geography		Unit 14: World Multipurpose River Projects		No. of periods: 14
Key unit competence: Evaluate the impact of multi-purpose river projects on sustainable development of the world.				
Learning Objectives			Content	Learning Activities
Knowledge and Understanding	Skills	Attitudes and Values		
– Give the meaning of multipurpose river projects. – Explain the aims and the importance of multipurpose river projects. – Identify the problems affecting the multipurpose river projects. – Identify the major multipurpose river projects of the world.	– Explain the importance of the multipurpose river projects. – Evaluate the problems affecting the multipurpose river projects and suggest possible solutions. – Examine the effects of the selected multipurpose river projects (Aswan High dam in Egypt, Huang He basin in China, Tennessee Valley Authority, Akosombo Dam Project) on the sustainable development of Egypt, Ghana, USA and China).	– Appreciate the aims and the importance of the multipurpose river projects in the development of the world. – Show concern for the problems affecting the multipurpose river projects and propose their possible solutions.	– Meaning Multipurpose River Projects – Aims and importance of Multipurpose River Projects – Problems affecting Multipurpose River Projects. – Problems affecting Multipurpose River Projects and their possible solutions.	– Use the geographical documents, internet and maps to explain the major multipurpose river projects, their aims, importance, problems affecting the multipurpose river projects and the solutions. In groups, discuss and compile the findings for a class presentation.

			Case studies: – Aswan High dam in Egypt – Huang He basin in China – TVA (Tennessee Valley Authority) – Akosombo Dam Project	– Use the geographical documents, maps, internet and photographs to explore the effects of multipurpose river projects and then in groups, discuss the findings for a class presentation.
Links to other subjects: <i>Physics: Power and energy production; Economics: Land use and resource utilisation</i>				
Assessment criteria: <i>The learners are able to critically evaluate the impact of multi-purpose river projects on sustainable development of the world.</i>				
Teaching and learning materials: <i>Geographical documents, tactile and talking materials, maps, photographs, internet/Jaws software and field study resources and local environment.</i>				

– Topic Area: Human and Economic Geography			Sub-Topic Area: Economic activities	
S6 Geography		Unit 15: Environmental Conservation and Tourism in the world.		No. of periods: 24
Key unit competence: Analyse the impact of the conservation of natural resources and tourism on sustainable development of the world.				
Learning Objectives			Content	Learning Activities
Knowledge and understanding	Skills	Attitudes and values		
– Give the meaning of the term environment. – Identify the various components of the environment. – State the factors for the environmental degradation. – Identify the consequences of the environmental degradation	– Describe the concept of the environment and its components. – Examine the various types of environmental degradation. – Explain the causes of the environmental degradation. – Explain the consequences of the environmental degradation and the methods for environmental conservation.	– Appreciate the various components of the environment. – Show continual desire to conserve and protect the environment. – Appreciate the importance of environmental conservation.	– Meaning and components environment – Factors of environmental degradation – Consequences of environmental degradation – Methods of Environmental Conservation – Importance of environmental conservation.	– Use the geographical documents, internet, maps and photographs to identify the components of the environment, factors and the effects of the environmental degradation, and the methods of environmental conservation. In groups, discuss and compile the findings for a class presentation.

– Identify the methods for the environmental conservation. – Give the importance of environmental conservation. – Identify the types, causes and effects of pollution – Identify the types of catastrophes. – Identify the causes and the effects of the catastrophes. – Identify the major tourist areas in the world. – State the factors affecting the tourism in the world.	– Justify the reasons for conserving our environment. – Explain the main categories of catastrophes and by providing relevant examples. – Evaluate the causes and the effects of the catastrophes and propose the solutions. – Locate the major tourist areas in the world. – Explain the factors affecting the development of tourism in the world. – Justify the importance of tourism in the sustainable development of the World.	– Show the concern for the effects of the catastrophes and propose their preventive measures. – Initiate the environmental clubs at the school to promote the methods for the environmental conservation and the awareness through planting trees, grass, and flowers – Appreciate the importance of tourism on the sustainable development.	Pollution: – Meaning and types of pollution – Causes, effects and solutions Catastrophes: – Natural catastrophes – Non- natural catastrophes/ calamities – Effects and solutions of catastrophes Tourism: – Meaning, major tourist areas of the world – Factors affecting the development of tourism in the world. – Importance and impact of tourism	– Use textbooks, internet, photographs and video clips to identify various types of pollution and catastrophes, their causes and effects. Then in groups, discuss and compile the findings for a class presentation. – Make research about major catastrophes that happened in the world over the last 5 years. Discuss the findings and compile in a report for a class presentation. – Use the geographical documents, internet, maps and photographs to identify the major tourist areas of the world, the factors affecting the development of tourism, the importance of tourism, the problems affecting the tourism in the world and the possible solutions. In groups, discuss and compile the findings for a class presentation.
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– Give the importance of the tourism on the economic development. – Identify the problems of tourism. – State the impact of tourism on the environment and the development of a country.	– Explain the problems of tourism and propose the possible solutions. – Evaluate the impact of tourism on the environment and the development of a country.	– Show the concern for the problems affecting tourism and suggest the possible solutions.	– Problems affecting tourism in the world and the prospects – Case studies on tourism (Switzerland, East African Countries, USA)	– Use the geographical documents, maps, internet and photographs to explain the tourism in the selected countries (Switzerland, East African Countries, USA). In groups, discuss and compile the findings for a class presentation.
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Links to other subjects: *Biology: Ecosystem and biodiversity.*

Assessment criteria: *The learners are able to critically analyse the impact of the conservation of natural resources and tourism on sustainable development of the world.*

Teaching and learning materials: *Maps, photographs, internet, geographical documents, textbooks, maps, local examples.*

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8. APPENDIX

8.1. SUBJECTS AND WEEKLY TIME ALLOCATION FOR SINGLE SHIFT IN UPPER SECONDARY-GENERAL EDUCATION (S4-S6)

Mathematics and Science Stream 2

Subject	Number of periods		
I. Major Core Subjects: Examinable at National Level	(Period= 40 minutes)		
	S4	S5	S6
1. Mathematics	7	7	7
2. Economics	7	7	7
3. Geography	7	7	7
4. Physics	7	7	7
II. Minor Core Subjects: Examinable at National Level			
5. English	3	3	3
6. French	3	3	3
7. Kinyarwanda	3	3	3

Sub-total 1	37	37	37
III. General Subjects examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and religious studies	3	3	3
11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300